Risk, salience, and the answers we observe

A new Uganda structural-development paper, a Ghana tax-morale experiment, and a measurement classic from Nigeria · three peer-reviewed economics-journal articles. The following article pages are edited reading excerpts, not complete replacements for the originals.

LITERATURE PULSE

The common thread is that measured behavior depends on the environment in which choices and answers are produced. Ugandan farmers appear unproductive partly because incomplete insurance makes safe crop portfolios privately sensible. In Ghana, making a familiar mobile-money tax salient shifts beliefs about the state's legitimacy and tax compliance, especially among people exposed to the levy. In Nigeria, replacing direct interviewer questions with an anonymous list changes measured violence prevalence and even correlations used to study empowerment. **Literature trend:** applied economics is moving past treating choices, attitudes, and survey answers as context-free data. Stronger papers explicitly model the constraint, randomize the information or measurement environment where possible, and separate behavioral responses from artifacts of elicitation. That makes design choices part of the economic mechanism rather than a footnote.

01 CURRENT / AGRICULTURAL RISK AND STRUCTURAL DEVELOPMENT · STARTS ON PAGE 2 · Journal of Development Economics · accepted September 2026 · peer reviewed · open access

Incomplete Markets, Crop Portfolios, and Agricultural Productivity

Why do poor farmers keep planting safer, lower-yield crops? Using five waves of Uganda's nationally representative panel, the paper documents a yield-risk trade-off and shows that poorer households tilt production toward low-yield crops while most farms mix both types. A heterogeneous-agent incomplete-markets model jointly disciplines crop choice, intermediate inputs, wealth, income volatility, and consumption smoothing. Completing agricultural financial markets raises calibrated productivity by 12.8%, consumption by 17.6%, and welfare by 17.4%; 15.8% of the productivity gain comes from reallocating existing inputs across farmers and crops. These are structural counterfactuals, not causal treatment effects, and depend on crop-risk measurement, functional form, calibration, and the no-borrowing/depreciating-asset representation. The clever insight is that apparently conservative crop portfolios can be privately rational insurance and socially costly at once.

02 CURRENT / PUBLIC FINANCE, MOBILE MONEY, AND POLITICAL ECONOMY · STARTS ON PAGE 17 · World Development 207 · November 2026 · peer reviewed · CC BY 4.0

Mobile Money and the Social Contract: Experimental Evidence from Ghana

Can taxing mobile money weaken the very state capacity the tax is meant to build? A preregistered survey experiment with 1,151 respondents in three Ghanaian regions randomly made the unpopular e-levy salient before measuring tax attitudes. The prime lowers perceived tax legitimacy and willingness to comply, while support for redistribution is largely unchanged. Effects are concentrated among remittance recipients and opposition voters, consistent with reciprocity and partisanship, though subgroup contrasts are suggestive rather than clean mechanism tests. The sample deliberately oversamples remittance recipients and is not nationally representative; outcomes are stated attitudes, and the treatment changes salience rather than the tax itself. The paper matters because digital fiscal capacity can generate a political backlash that pushes citizens back toward cash and further erodes the social contract.

03 CLASSIC / INTERVIEW PRIVACY, MISREPORTING, AND VIOLENCE · STARTS ON PAGE 27 · World Bank Economic Review 37(1) · 2023 · peer reviewed classic · CC BY 4.0

Method Matters: The Underreporting of Intimate Partner Violence

How much does another person's presence change what a respondent reports? In rural Nigeria, women were randomly assigned either direct face-to-face questions or an anonymous list experiment about intimate-partner violence. Emotional and physical IPV estimates are 31% and 35% higher under the list method; physical IPV is 180% higher than the Nigerian DHS benchmark. Misreporting is systematic: vulnerability, relationship quality, literacy, and assets can have different or even opposite associations depending on interview method. Robustness checks find no detectable list-design effect and estimates survive enumerator fixed effects and alternative controls. List estimates are noisy, the no-liars assumption cannot be tested, and results vary by violence type and context. This is a direct warning that interviewer presence can bias both prevalence and estimated program effects.

Editorial note: references and appendices were omitted where possible; no included article contributes more than about 15 text pages, excluding table-heavy pages. Full citations and source links remain in the catalog. AI-written synopses were checked against the included source pages.

Incomplete Markets, Crop Portfolios, and Agricultural Productivity

ALBERT RODRÍGUEZ-SALA*
Universidad de Cantabria
August, 2026

Using the Uganda National Panel Survey, I document that farmers face a trade-off between crop yields and risk, that most grow both high- and low-yield crops, and that poorer farmers concentrate on the low-yield, low-risk ones. I then develop an incomplete-markets model with a crop-portfolio choice, in which households trade off income returns against consumption smoothing. Wealth-poor households optimally under-invest in intermediates and shift toward safer crops. Crop portfolios, intermediate input use, and the wealth distribution are jointly determined in equilibrium. The model also reproduces the low pass-through of both income inequality and income volatility into consumption. Calibrated to rural Uganda, it implies that completing markets raises agricultural productivity by 12.8%, with 15.8% of the gain coming from eliminating the misallocation of inputs across farmers and crops. Consumption rises by 17.6% and welfare by 17.4%, with income inequality rising slightly and consumption inequality unchanged.

Keywords: Agriculture, Risk, Incomplete markets, Crop choice.

JEL Classification: O11, 013, E21.

Households in low-income countries live with high levels of risk while having limited access to formal financial markets or public safety nets. Many of these households rely heavily on agriculture, a sector characterized by both high income risk and particularly low productivity (Caselli, 2005; Restuccia, Yang, and Zhu, 2008; Gollin, Lagakos, and Waugh, 2014). In this article, I study how risk distorts agricultural decisions among heterogeneous farmers to then quantify the aggregate impact of completing financial markets in agriculture in Uganda.

Complete markets tests in developing countries typically show a high degree of consump-

*I thank the editor and three anonymous referees for comments that improved the paper. This paper is based on my PhD dissertation at Universitat Autònoma de Barcelona, and I thank the IDEA-UAB program and, above all, my advisor, Raül Santaeulàlia-Llopis. I also thank Douglas Gollin, Kevin Donovan, Ethan Ligon, Jan Grobovšek, Jordi Caballé, Alex Ludwig, Virginia Sánchez-Marcos, Marc Teignier, Vahagn Jerbashian, and Luis Rojas for their comments, and seminar participants at UAB, Toulouse School of Economics, University of Edinburgh, Universitat de Barcelona, SAEe 2021, CESC 2023, AFES 2024, EWMES 2025, and STEG 2026. I acknowledge financial support from the Generalitat de Catalunya under grant 2021 SGR 00194 and the Spanish Government grant PID2024-158095NB-I00.

tion insurance (Townsend, 1994).[1] Yet informal insurance mechanisms are often imperfect and costly. When farmers in developing countries are offered insurance products, they tend to take riskier and more productive decisions, such as increasing fertilizer usage (Karlan, Osei, et al., 2014) or shifting to more productive crops (Mobarak and Rosenzweig, 2013; Cai, 2016). The absence of perfect insurance can have large aggregate effects, though knowledge on this dimension is still limited. Donovan (2021) shows how uninsurable risk reduces the intermediate share and in equilibrium creates a large agricultural productivity gap.

I use data from the Uganda National Panel Survey (UNPS), which provides detailed agricultural information at the household-plot level. I measure yields as the value of production—including non-sold output valued at local median consumption prices—per land unit ($/acre). Exploring the relationship between crop yields, risk, and crop choice, I find three empirical results. First, there is a strong positive correlation between crop yields and risk: crops such as plantain bananas, rice, and groundnuts offer high but volatile yields, while beans, cassava, and sorghum offer lower and less volatile yields. Second, the share of output from high-yield, high-risk crops rises along the wealth distribution: the poorest farmers obtain most of their production from low-yield crops (mainly beans, maize, and cassava), the richest mainly from high-yield crops (mainly plantain bananas). Third, most farmers cultivate high- and low-yield crops simultaneously. These patterns are robust across alternative risk measures, crop classifications, and data sources.

Motivated by these results, I develop an incomplete-markets model with a crop-portfolio choice. The economy is agrarian, populated by household farms that differ ex ante only in a permanent component of agricultural productivity. Households are infinitely lived with standard CRRA preferences, and income is endogenous to farming decisions: they choose intermediate inputs for both the high-return, high-risk technologies (high-yield crops) and the low-return, low-risk ones (low-yield crops), and also receive non-agricultural earnings from an exogenous stochastic process.[2] Financial markets are incomplete, in the tradition of Bewley (1986); Imrohoroğlu (1989); Huggett (1993); Aiyagari (1994); to capture the hardship of saving in developing countries, and following Donovan (2021) and Lagakos, Mobarak, and Waugh (2023), saving occurs through a risk-free asset that depreciates, subject to a no-borrowing constraint.[3] Because most agricultural production in Uganda is at subsistence level and rural labor markets are thin, I model the economy as a small open economy with fixed input prices and

[1] Townsend (1994) finds that household consumption responds little to idiosyncratic income shocks in rural India. Similarly, Paxson (1993) finds limited evidence that consumption tracks income seasonality in Thailand. More recent studies with results in the same direction include Chiappori et al. (2014), De Magalhães and Santaeulàlia-Llopis (2018), among others.

[2] Adamopoulos and Restuccia (2020) also model crop selection as a choice between two technologies, where the key trade-off is higher productivity versus lower fixed costs: a "cash-crop" technology and a "food-crop" technology.

[3] Although risk-sharing is a critical source of insurance in developing countries (Udry, 1994; Ligon, 1998; Fafchamps and Lund, 2003), for tractability insurance in the model is limited to self-insurance. Quantitatively, the model still captures the aggregate degree of insurance in the data—measured by the transmission of income to consumption volatility.

work in partial equilibrium.[4]

The model features two interlinked risk-management mechanisms: precautionary saving and precautionary income choices. When choosing intermediate inputs, households trade off higher agricultural returns against less consumption smoothing, and they adjust along two margins. The first is the total amount of intermediate use: holding the portfolio fixed, households raise returns by using more intermediates, and if only this margin operates there is a monotonic relationship between intermediate use, returns, and risk. The second margin is the portfolio choice across technologies: households can raise returns by shifting from low- to high-return technologies without increasing total intermediate use. Formally, this trade-off appears in the Euler conditions for intermediates, where a risk-induced wedge varies by technology. Portfolio choice thus adds an allocation problem within the household farm, so that the distribution of risk across technologies can generate misallocation. This microfounds the aggregate risk-induced wedge that an intermediate-use problem without a crop portfolio would otherwise leave unexplained.

Using the UNPS data, I calibrate the model to rural Uganda. Part of the ISA-LSMS project, the survey measures agriculture, consumption, income, and wealth within the same households, letting me reconstruct the entire budget constraint from a single dataset. Some parameters are set externally—the discount factor and risk aversion from the literature, and the crop-risk and non-agricultural income processes estimated directly from the data—while the remaining ones, the two technology factors, the dispersion of permanent productivity, and the depreciating return on saving, are calibrated within the model by a near-triangular method of moments, with each pinned mainly by one moment of the UNPS.

Beyond the moments it targets, the calibrated model reproduces the slope of the second empirical fact documented above: the rising share of high-yield, high-risk crops along the wealth distribution. It also delivers predictions for the joint distribution of consumption, income, and wealth. Because these are endogenous in an incomplete-markets economy, this class of models provide a theory of their joint distribution—used mostly to account for inequality in the United States and other high-income economies (Castañeda, Díaz-Giménez, and Ríos-Rull, 2003; Krueger and Perri, 2006). For Sub-Saharan Africa, De Magalhães and Santaeulàlia-Llopis (2018) document two features of the region: a low transmission of income inequality to wealth inequality (low accumulation in the life cycle), and a low transmission of income inequality to consumption inequality (high consumption insurance). Yet, to my knowledge, no structural account of these distributions exists for the region. This paper provides one. Although the calibration targets no dispersion moment, the model reproduces the second fact, high-insurance, in both dimensions in which it appears in the data: a cross-sectional one—the low pass-through of income inequality to consumption inequality—and a dynamic one—the low pass-through of idiosyncratic income volatility to consumption volatility. It thus captures quantitatively the high consumption insurance observed in rural developing economies and matches the

[4] This abstracts from the labor-reallocation channel Donovan (2021) finds important in India—which would raise the gains—and from local relative-price adjustment—which could lower them.

broader consumption–income distribution of rural Uganda with standard preferences and no consumption target, though it under-predicts the concentration of liquid wealth.This suggests that the incomplete-markets toolkit—the workhorse for inequality and welfare analysis in rich economies—is a useful and underused framework in developing economies.

Next, I develop a planner's problem that replicates the competitive-equilibrium allocations of intermediates and consumption under complete markets in agriculture, where households maximize discounted expected profits through their crop-portfolio choices. Comparing the calibrated benchmark with this counterfactual delivers the main results.

Completing financial markets raises agricultural productivity by 12.8%; because labor and land are fixed, output and productivity—per worker and per land unit—rise together. The gain comes from two channels: a 29.0% rise in average intermediate use, and a more efficient allocation of intermediates across the crop portfolios of heterogeneous farmers. Holding aggregate intermediate use at its benchmark level and allocating it efficiently accounts for 15.8% of the total increase in production, 12.2% across farmers and 3.2% across crops. The crop portfolio itself accounts for 16.0% of the gain: an economy with a single technology, recalibrated to the same average agricultural output and the same volatility of total agricultural output, delivers a productivity gain of 10.7% against 12.8% here. Completing markets also lowers the demand for costly precautionary saving which, together with the productivity gain, raises aggregate consumption by 17.6%.

In this model, the effect on consumption inequality is theoretically ambiguous: completing markets removes the dependence of consumption on agricultural shocks (an egalitarian force), but the largest output gains accrue to high-productivity households concentrated in the middle and upper distribution, and the reduction in costly saving benefits the top most. After completing markets, income inequality rises while consumption inequality remains almost unchanged, as the egalitarian and inequality-enhancing forces roughly offset.[5] In consumption-equivalent terms, households would need a 17.4% increase in lifetime consumption to be indifferent between being born into the incomplete-markets economy and the economy with complete agricultural markets.

Related Literature

The closest paper is Donovan (2021), who shows that, in equilibrium, uninsurable risk lowers the intermediate share, so that completing financial markets raises agricultural productivity and narrows the cross-country productivity gap. I contribute in three ways. First, risk distorts intermediate use along two margins: its level, as in his model, and its allocation across crops, which in turn feeds back on the level. Second, I quantify the welfare gains from completing agricultural insurance: the gains in aggregate consumption are larger than in output, 17.6% against 12.8%, and the distributional effects are non-trivial. Third, in a poorer,

[5]Households with low permanent components—limited agricultural skills, small landholdings, or remote, unproductive locations—may gain little from insurance.

subsistence-based economy and under different assumptions—a permanent productivity component, no subsistence requirement, and no cross-sector labor reallocation—the core insight survives: uninsurable risk lowers agricultural productivity.[6]

More broadly, the paper contributes to two strands of work on agricultural productivity in poor countries: a macro-development literature on misallocation, and a micro literature on risk in farm decisions. The first documents the large productivity gap between rich and poor countries and the barriers behind it: high transport costs and subsistence (Gollin and Rogerson, 2014), land-market frictions under communal tenure (Gottlieb and Grobovšek, 2019; Adamopoulos and Restuccia, 2020; Chen, Restuccia, and Santaeulàlia-Llopis, 2023), land and financial frictions (Manysheva, 2022), land-use restrictions that distort crop choice (Le, 2020), among others. In that literature, misallocation operates largely across farmers, in the factors they use (Adamopoulos and Restuccia, 2014). I add the margin misallocation across crops, within the farm. It connects to Adamopoulos and Restuccia (2022), who show that the productivity gap reflects economic choices—such as which crops are grown, and how—rather than unfavorable geography; I show that part of that inefficient crop allocation stems from optimal choices of farmers facing uninsurable risk. The second examines the role of risk in agricultural decisions: Dercon (1996) shows that risk distorts crop choices, Dercon and Christiaensen (2011) that it distorts fertilizer use, and Rosenzweig and Binswanger (1993) that its effect on investment depends on wealth, while Kurosaki and Fafchamps (2002) find that crop choices respond to risk even when accounting for village-level insurance. The contribution of this paper is twofold: it develops a framework that incorporates these micro-level mechanisms into an equilibrium setting with an endogenous wealth distribution, and it quantifies the aggregate consequences of them for agricultural productivity and welfare. Finally, by quantifying the productivity and welfare cost of the distortions households accept to smooth consumption, the paper speaks to work documenting a trade-off between consumption smoothing and growth, at the micro (Rosenzweig and Wolpin, 1993; Morduch, 1995) and macro (Munshi and Rosenzweig, 2016; Santaeulalia-Llopis and Zheng, 2018) levels.

1 Empirical Findings

This section explores the relationship between agricultural yields, risk, and farmers' crop choices across the main crops grown in Uganda. I document three facts: (1) there is a positive yield–risk tradeoff across crops, robust across alternative risk measures and data sources; (2) low-yield, low-risk crops are the main share of output for poorer farmers; and (3) most farmers grow both low- and high-yield crops at the same time.

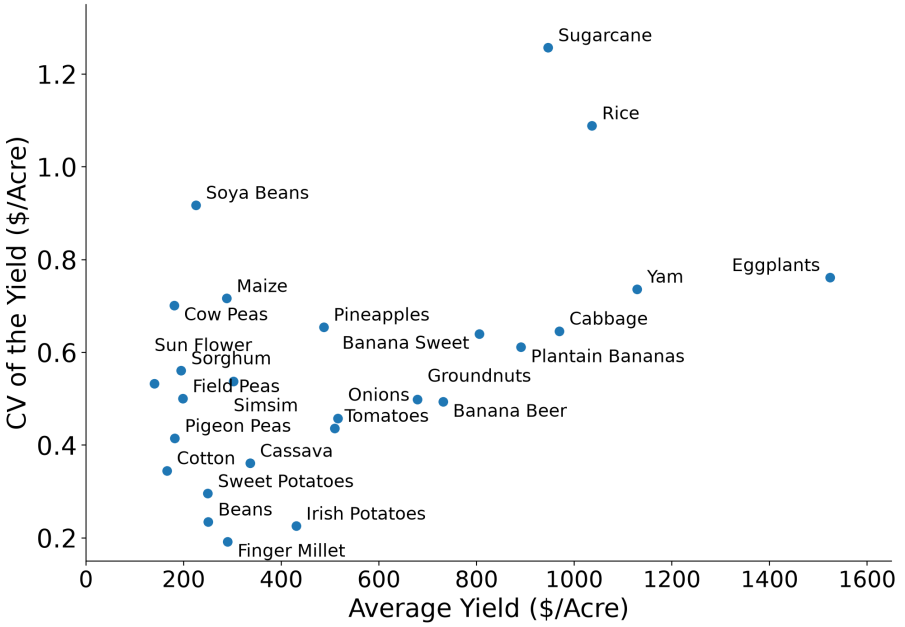
To document these facts, I mainly use the first five waves of the Uganda National Panel

[6] Donovan (2021) studies India with a two-sector model (agriculture and a riskless industry); completing markets there raises productivity through a higher intermediate share, reduced misallocation across farmers, and labor reallocation to industry. I omit the third channel, which requires an industrial sector to absorb post-shock farm labor—hard to justify in rural Uganda.

Survey (UNPS) and compute crop yields, defined as the value of production per land unit ($/acre). The value of production is the household-reported value of crop sales plus non-sold production, valued at district-level median consumption prices, and land is the reported acreage devoted to the crop.

Fact 1: a yield–risk tradeoff. Figure 1 plots average monetary yield against the coefficient of variation (CV) of yield across crops, with observations aggregated at the wave–season level so that dispersion reflects variation over time. Higher-yielding crops are markedly more volatile. Crops such as plantain bananas (matooke) reach average yields of about $891 per acre with a CV of 0.61, whereas crops such as beans yield around $250 per acre with a CV of 0.24—a threefold difference in yields and a two-and-a-half-fold difference in risk. The positive relationship is strong with a correlation of 0.68 and holds across alternative risk measures and data sources—which I discuss in section 1.1.

Figure 1: Crop Yield and Crop Risk in Uganda

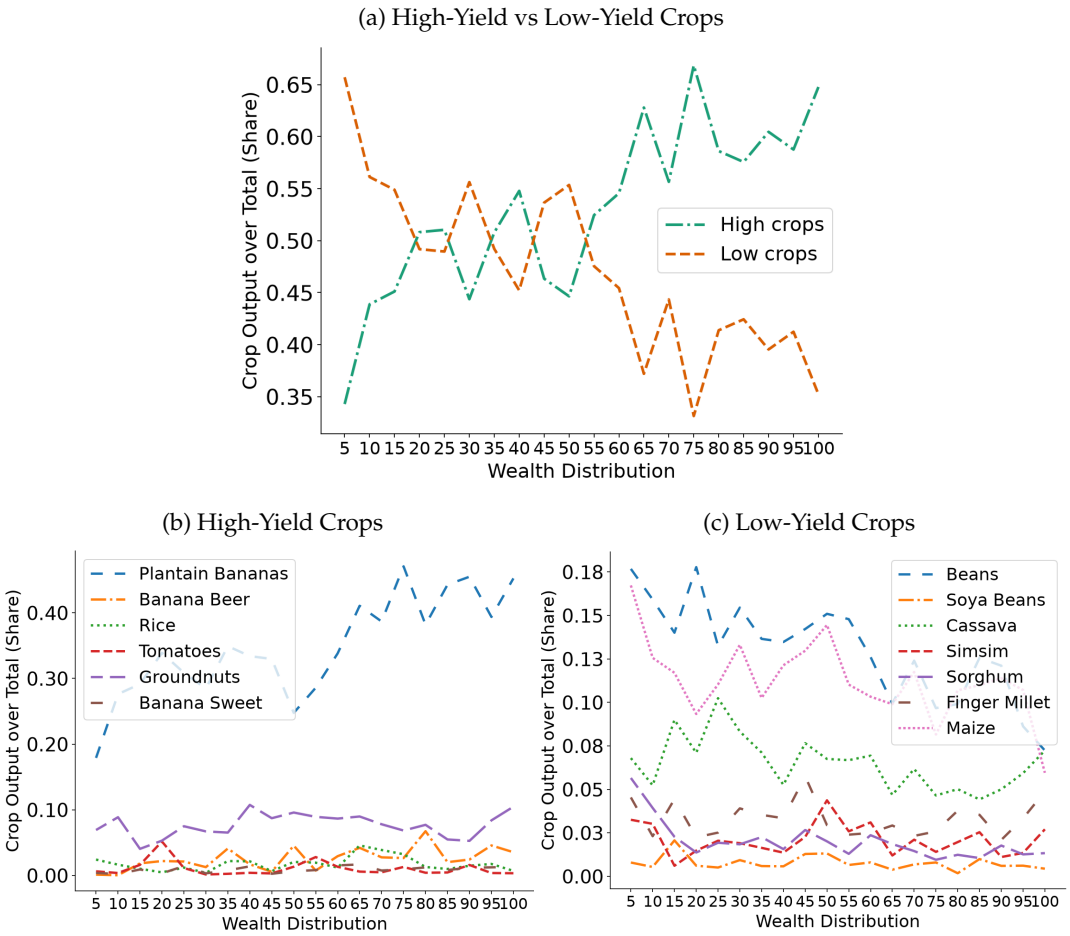


Notes: Yield observations ($/acre) at the wave level. Crops with a planting-to-harvest cycle under two years. *Source:* UNPS 2009/10–2015/16.

Fact 2: poorer farmers concentrate output in low-yield crops. I group the non-long-term crops into high- and low-yield categories by whether their average yield is above or below the median crop. High-yield crops include bananas (beer), plantain bananas, sweet bananas, cabbage, eggplants, groundnuts, Irish potatoes, onions, pineapples, pumpkins, rice, sugarcane, tomatoes, and yam; low-yield crops include beans, cassava, cotton, dodo, cowpeas, field peas, finger millet, maize, pigeon peas, simsim, sorghum, soya beans, sunflower, and sweet potatoes. Figure 2 shows the crop portfolio along the wealth distribution. Low-yield crops make up the majority of poorer farmers' output and high-yield crops the majority of richer farmers. Among the poorest farmers (0–10th percentiles) low-yield crops are around 60% of production; in the

middle (15th–55th) output is split roughly evenly; and at the top (60th percentile and above) high-yield crops account for about 60%.

Figure 2: Crops Production along the Wealth Distribution



Notes: Wealth includes estimated land value, household assets, livestock, and farming capital. Crop production and wealth ranking computed from the average across the five waves.
Source: UNPS 2009/10–2015/16.

Sub-plots (2b) and (2c) show the six most cultivated high-yield and seven most cultivated low-yield crops. The rise in the high-yield share is driven mainly by plantain bananas, which go from 33% of total production for the bottom 50% to 40% for the top 35%; conversely, beans and maize together fall from about 30% of the bottom 50% poor's output to under 20% for the rich. The same gradient appears more strongly along the income distribution (Figure 6 in the Appendix).

Fact 3: farmers grow both types. Most farmers grow both low- and high-yield crops. Table 1 reports the share of households growing low-yield crops, high-yield crops, and both, across waves. On average 65.0% grew high-yield crops and 96.2% grew low-yield crops; over the whole panel, 98.6% grew low-yield crops, 76.1% grew high-yield crops, and 75.3% grew both.

Perennial crops and banana risk. The analysis retains crops with a planting-to-harvest cycle under two years and excludes long-gestation perennials (coffee, tea, tobacco, and others), whose yields and volatility are hard to measure and whose fixed costs are a margin this paper does not study. Among the retained perennial crops, only bananas are large enough to shape the aggregate patterns: they are the high-yield crop that accounts for most of the wealth gradient in Fact 2. Because bananas are perennial, one might worry that their measured volatility reflects the maturation of young plants rather than genuine production risk. I check this by comparing growers first observed cultivating bananas in the early waves ("established") with later adopters.[9] If maturation were inflating banana risk, established growers would show lower volatility. They do not: over the common window their within-household CV is 0.79, slightly above later adopters (0.71) and the cross-crop median (0.72) while over the full panel the gap is wider, 0.88 against 0.71 (Appendix Table 11). Bananas are also a staple grown by many Ugandan farmers rather than a specialized crop and recalibrating the model without bananas delivers gains close to those of the baseline classification.

Alternative crop classifications. The trade-off between high- and low-return crops is preserved across alternative classifications. Appendix Table 12 reruns it five ways: keeping only crops with more than 150 growers, cutting at terciles rather than the median, excluding bananas, using the GAEZ physical ranking, and grouping crops and measuring their risk at the district level. In every case the group of high yield crops exhibits higher risk than the low yield one. Across the value-based splits the variance ratio stays around its baseline value of 1.72 (from 1.69 to 1.79), rises to 2.10 when crops are grouped at the district level, while it decreases to 1.08 in the GAEZ split. This is expected since GAEZ ranks crops by agronomic physical potential (attainable yield), not by return; and because it contains no time variation, the risk it is paired with comes from the UNPS panel. Finally, The two-group structure is not discarding much of the crop-level variation in risk. Partitioning the same crops into three or five groups raises the average within-group output risk by only $3-8\%$ and leaves the spread of risk across groups essentially unchanged.

2 Model

2.1 Primitives of the Model

The model depicts an agrarian economy populated by a continuum of household farms—or simply households—of measure one facing idiosyncratic risk with no aggregate risk. They are ex-ante identical except for a permanent component in agricultural production.

than three percentage points. The small negative covariance is the natural-insurance association of Allen and Atkin (2022), concentrated at the market rather than the idiosyncratic level (-0.65 across waves versus -0.06 to -0.08 idiosyncratically). Appendix Tables 9 and 10 report the full results.

[9] Plant age is not observed, so I use a proxy: farmers first observed cultivating bananas in 2009/10 or 2010/11 ("established") plausibly have fewer young plants than farmers first observed growing cultivating from 2011/12 onward. I compute the risk measures for both groups over the common window 2011/12–2015/16, in which both are observed, so the comparison spans the same years.

3.1 Consumption

The UNPS includes three sections for household consumption: food consumption (over the last seven days), non-food and non-durables consumption (over the last month), and durables consumption (over the last year). I compute the consumption value for each good or service by summing the values from purchases, home production, and in-kind and gift transfers. Then, I compute the household aggregate consumption as the sum of food and non-durable items consumption, excluding durable goods. To maintain data integrity, outliers are trimmed by excluding the bottom and top one percent of total household consumption for each wave.

3.2 Income

Measuring household income in developing countries is challenging; see Deaton (2005) and Deaton (2019) for useful discussions. First, a significant portion of the economy operates within the informal sector, leading to irregular periods of work and frequent in-kind payments. Second, most households are self-employed, making it challenging for them to report their earnings accurately both conceptually and practically. Third, many of these self-employed individuals are farmers, which presents further data measurement issues, especially when many farmers operate at the subsistence level. To overcome these challenges, the LSMS-ISA surveys query households about earnings derived from formal and informal labor services, non-agricultural business and trade operations revenues, and costs (LSMS section). Moreover, the LSMS-ISA surveys incorporate an exceptionally detailed agriculture questionnaire (ISA section).

3.2.1 Agricultural Income

With the agriculture questionnaire, I compute the agricultural revenues for each household and each crop by summing the earnings from sales plus the value of non-sold production, using district-level median consumption prices per crop kilogram. From the production side, households are questioned about the total production of the crop, as well as the production of the crop allocated to the market, own consumption, gifts to other households, storage, and animal feed. Like many surveys in developing countries, households can report crop production in various units, including non-standard units such as baskets, pails, or bunches. Additionally, conversion rates can vary across crops and seasons. For accurate conversion of crop production into kilograms, I use the median of the conversion rates to kilograms reported by households for each wave, season, crop, and unit combination.

Subsistence farming is prevalent in Uganda. On average, households devote only about 26% of their agricultural output to the market. Thus, understanding Ugandan households' livelihoods and measuring agricultural yields requires accounting for non-sold agricultural production correctly. However, as Deaton (2019) argues, setting prices to evaluate non-sold production in monetary terms is challenging. While prices at the gate (selling prices) are commonly used in the literature, they may underestimate non-sold production. This is because

the items sold may only represent a portion of the harvested crops, and prices at the gate are typically measured after the harvest when supply is abundant and prices are low.

To address this issue, I leverage the household questionnaire, which captures food consumption in quantity and value. Following the approach in De Magalhães and Santaeulàlia-Llopis (2018), I use median consumption prices at the district level to evaluate unsold agricultural production. This procedure allows for better capturing the shadow value of not selling the crop: the cost if the household did not produce the crop and had to buy it in the local market. Agricultural revenues for each season and wave are typically trimmed at the top 2 percent of the distribution.[16]

3.2.2 Intermediates

Agricultural intermediates expenditure is computed as the sum of the costs of chemical fertilizer, organic fertilizers, pesticides and herbicides, seeds, hired labor, and transport. Since the data report the use of intermediates at the plot level, for multi-crop plots I assume that the intermediates are distributed uniformly across the crops on the plot. For chemical fertilizer, organic fertilizer, pesticides, and herbicides, the costs include total expenditure plus the value of the non-bought quantities using median prices recovered from the survey.

Agriculture in Uganda is characterized by a minimal usage of intermediates. Table 14 in the Appendix presents the share of farmers using intermediates and their average expenditure across the five waves of the UNPS. On average, farmers spend only $78.14 per year on intermediates. Decomposing across crops, the average intermediates expenditure in high-yield, high-risk crops is $57.76, while for low-yield, low-risk crops it is $46.25. The use of modern agricultural intermediates in Uganda is extremely low, with only 4.7% of farmers using chemical fertilizer and 12.8% using pesticides.

3.2.3 Non-Agricultural Earnings

Non-agricultural earnings include business profits, formal and informal labor income, and livestock earnings. The household questionnaire collects detailed data on each non-farming business, including operative months, gross revenues, wage expenditures, raw materials costs, and other expenditures. Business profits are calculated as the difference between monthly gross revenues and costs, multiplied by the number of months the business operated. Similarly, labor earnings aggregate yearly payments for primary and potential secondary jobs for each household member, regardless of whether the job is formal or informal. Livestock profits are determined by summing net animal sales, meat, milk, and egg sales, and subtracting associated costs. The value of unsold production is estimated based on the consumption of self-produced livestock products reported in the household questionnaire's consumption section.

[16] Uganda has 135 districts. For crops without a district-level consumption price, such as sorghum and groundnuts, I use district-level selling median prices. If these prices are unavailable, I use regional-level prices, and if those are also unavailable, I use nationwide prices.

3.3 Wealth

The data equivalent of assets in the model is liquid wealth, which encompasses stored crops, livestock, household assets, rents, and transfers. Household assets include a wide range of items such as housing and other buildings, non-agricultural land, furniture, household appliances, electronic devices, jewelry, vehicles, and other assets. Rents and transfers consist of property income, dividends, interest from current accounts or bonds, remittances, inheritance, and alimony.

The value of stored crops is determined using district-level consumption prices, while the values of livestock and household assets are based on self-reported values if the household were to sell the asset at the current moment. While in the calibration wealth only includes liquid assets, for the empirical analysis, wealth encompasses all assets owned by the household, including also farm capital and land value.[17]

3.4 Consumption, Income, and Wealth Summary

Table 16 in the Appendix summarizes household consumption, income, and wealth for rural Uganda (columns 2-4) and the entire country (columns 5-7), highlighting three insights. First, income and consumption estimates closely align, indicating high data quality. Income estimates from surveys in developing countries typically lag behind consumption estimates for various reasons including those related to the difficulties of measuring agricultural income—see (Deaton, 2005). Second, Uganda remains a very poor economy. In rural areas, average household income and consumption are around $1,505 annually, rising to $1,736 nationwide. On a per capita basis, this equates to $287 in rural areas and $338 nationally. Third, while consumption inequality is comparatively low (Gini index of 0.33 respectively) income inequality levels are high with a Gini index of 0.57—compared to the United States. [18]

4 Calibration

A set of parameters is taken from the literature or estimated directly from the UNPS data; the remaining ones—the technology factors A and B, the intermediates price p, the productivity dispersion σ_z, and the depreciation rate δ—are calibrated within the model so that its stationary equilibrium reproduces a set of time-averaged moments in rural Uganda. The calibration is a near-triangular method of moments: each calibrated parameter is pinned mainly by a single moment, while average income and the dispersion of wealth are left as over-identifying checks.

[17] To value land, in the first two waves, households reported their value, allowing for the computation of per-acre land prices grouped by counties and plot characteristics. For the other waves, land value is estimated by applying the county per-acre prices from 2010–11 according to plot characteristics.

[18] Values in 2013 US$ not adjusted by PPP. Using 2006 PSID statistics, the Gini index of consumption in the United States is 0.41, while the Gini index in income and in wealth are 0.44 and 0.61, respectively—see De Magalhães and Santaeulàlia-Llopis (2018). Note that the UNPS has a relatively small sample and it does not over-sample the rich.

5 Complete Markets in Agriculture: the Planner's Problem

I obtain the main quantitative results of this paper by comparing the previous benchmark economy to an economy with complete financial markets on agriculture. To solve this counterfactual world, I develop the following social planner's problem.[21]

The planner is able to allocate intermediates and consumption allocations from the total agricultural production. To formulate the problem, I follow the approach in L. Maliar and S. Maliar (2003), and I represent the planner's problem in the form of two sub-problems. The first sub-problem is to distribute aggregate consumption across heterogeneous agents, which delivers the social period utility

$$U(C_t) = \underset{c_{it}}{\text{Max}} \int_i \omega_i u(c_{it}) di \tag{20}$$

$$\text{subject to} \qquad \int_i c_{it} di = C_t \tag{21}$$

in which ω_i is the Pareto weight on individual i. The second sub-problem is to solve for the aggregate consumption that maximizes the social preferences

$$\underset{\{C_t, m^h_{it}, m^l_{it}\}_{t=0}^{\infty}}{\text{Max}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t U(C_t) \tag{22}$$

$$\text{subject to} \quad C_t = \int_i \left[\theta_{it} z_i A (m^h_{it-1})^{\alpha} + \varepsilon_{it} z_i B (m^l_{it-1})^{\gamma}\right] di - p \int_i \left[m^h_{it} + m^l_{it}\right] di \tag{23}$$

$$C_t = Y_t - pM_t \tag{24}$$

in which aggregate consumption at period t is fully determined by the investment decisions on intermediates across individuals and technologies. Hence, the first-order conditions with respect to m^h_{it} and m^l_{it} are

$$\left[m^h_{it}\right] : \mathbb{E}_t \left[\beta^{t+1} U_c(C_{t+1}) \alpha \theta_{it+1} z_i A (m^h_{it})^{\alpha-1}\right] = \beta^t u_c(C_t) p \tag{25}$$

$$\left[m^l_{it}\right] : \mathbb{E} \left[\beta^{t+1} U_c(C_{t+1}) \gamma \varepsilon_{it+1} z_i B (m^l_{it})^{\gamma-1}\right] = \beta^t u_c(C_t) p \tag{26}$$

for all individuals $i \in \mathcal{I}$ at period of time $t > 0$. Given that θ_{it+1} and ε_{it+1} are independently distributed across time and unobservable by the planner, the intermediates allocations m^h_{it}, m^l_{it} are also independent of the idiosyncratic shocks realizations θ_{it+1} and ε_{it+1}. Also, note that in this economy, there are no aggregate shocks, and households have preferences for a smoothed consumption profile, so aggregate consumption growth needs to be constant. Thus, the expec-

[21] To replicate complete markets in agriculture, the planner provides full insurance for the agricultural output, but the planner cannot observe nor dictate on the realizations of the non-agricultural earnings.

6.1.2 Risk-Induced Misallocation

To measure the cost of the misallocation of inputs across farmers and crops, I constrain the complete-markets solution to the benchmark's level of investment. I use the planner's framework to hold the investment shares across the crop portfolio and among heterogeneous farmers at their complete-markets values, with aggregate intermediate expenditure matching the benchmark economy. Holding those shares fixed equalizes the marginal products across crops and among farmers with different permanent components, satisfying the optimality conditions in (33) and (32). The counterfactual is therefore the most output obtainable from the inputs the economy already uses.

Reallocating both margins accounts for 15.8% of the total productivity gain; equivalently, risk-induced misallocation costs 2.1% of agricultural output. Reallocating across farmers alone accounts for 12.2% and across crops alone for 3.2%, the residual 0.4% being their interaction.

6.2 Consumption and Welfare

In the absence of complete financial markets, households mitigate agricultural risk through income choices and costly precautionary savings. Completing markets removes the need for both, producing a substantially larger impact on aggregate consumption than on income. Panel A of Table 6 compares household averages of agricultural output, income, assets, and consumption across the two economies. Completing markets raises agricultural output by 12.8%, which translates into an 8.6% rise in household income. Without complete markets, households accumulate wealth through costly means—livestock, crop storage, or hoarding cash; completing markets reduces these costly assets by 59%. Combined with the higher income, this delivers a 17.6% increase in aggregate consumption.

Panel B of Table 6 shows the impact on the income and consumption distributions: the percentage change for households in the bottom of the distribution (bottom 1%, 5%, 10%, 25%, 50%) and for the same groups at the top. Completing markets has theoretically ambiguous effects on inequality. Insurance is egalitarian: under complete markets consumption no longer depends on shock realizations, so—conditional on the permanent component—households with good or bad agricultural luck consume the same. Working against this are two forces: households with high permanent components z_i gain the most in output and are concentrated in the middle and top of the distribution, and the reduction in costly savings benefits the asset-rich top disproportionately.

For income, the inequality-enhancing forces dominate: the income Gini rises from 0.53 to 0.54, with the bottom 50% gaining 7.8% and the top 50% gaining 9.1%. For consumption, the egalitarian force roughly offsets them, leaving inequality essentially unchanged—the consumption Gini varies from 0.371 to 0.370. The consumption gains are large throughout but not monotone: they are largest in the lower tail, where insurance raises the worst outcomes (bot-

so the two technologies capture most of the scope for diversification.

tom 1% +26.6%, bottom 5% +17.9%), sizeable through the upper-middle (top 50% +18.3%, top 25% +17.9%), and smallest at the very top (top 1% +2.4%).

Table 6: Consumption and Welfare Gains from Completing Markets

A. Aggregate Gains	Incomplete Markets (Income = 100)	Complete Markets (Income = 100)	$\triangle_{CM}$ (%)
Agric. Output	66.7	75.3	12.8
Income	100.0	108.6	8.6
L. Wealth	121.8	49.8	-59.1
Consumption	63.1	74.2	17.6

B. Distributional Gains	Bottom					Top				
	1	**5**	**10**	**25**	**50**	**50**	**25**	**10**	**5**	**1**
			(%)					(%)		
Income	3.7	3.1	4.9	3.8	7.8	9.1	9.5	10.2	6.5	13.9
Consumption	26.6	17.9	16.8	14.8	15.5	18.3	17.9	10.0	6.2	2.4

Notes: Panel A normalizes all quantities by aggregate incomplete-markets income (set to 100); the aggregate resource constraint is $C = Y_a - pM + Y_{na} - \delta A$. Panel B reports the proportional change in total income (row 1) and consumption (row 2) of the bottom and top percentile groups. The income Gini rises from 0.53 to 0.54; the consumption Gini is essentially unchanged, from 0.371 to 0.370.

Consider the ex-ante welfare comparison of the benchmark and complete-markets economies. The average welfare gain, in consumption-equivalent variation (CEV), is the constant proportional increase in consumption g, common across time and agents, that equates utilitarian welfare across the two economies:

$$\int \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t u((1+g)c_{it}^{IM}) \right] d\lambda^* = \int \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t u(c_{it}^{CM}) \right] d\lambda^*. \tag{36}$$

Under the baseline calibration, the welfare gain g is equivalent to a 17.4% increase in lifetime consumption. The welfare gain is almost equal to the gain in aggregate consumption because the upper half of the distribution gains slightly more than the lower half, pushing the welfare gain below the consumption gain; while the fall in within-household consumption volatility pushes it up. Neither is large: the consumption Gini moves only from 0.371 to 0.370. The welfare gain mainly reflects the level of consumption rather than a change in how it is distributed or in how much it fluctuates.[25]

[25]Completing markets lowers the within-household variance of log consumption from 0.110 to 0.072, but raises the variance between households from 0.407 to 0.451: consumption no longer moves with agricultural shocks, but it depends fully on the permanent component. Cross-sectional dispersion is therefore almost unchanged. Because the welfare criterion below aggregates utilities across households and dates, it responds to that cross-sectional dispersion rather than to the within-household component separately, which is why the welfare gain tracks the rise in aggregate consumption.

6.3 Discussion and Robustness

Are these gains large? Set against the large cross-country agricultural productivity gaps and the sizeable misallocation shares documented in the macro-development literature, a 12.8% productivity gain is moderate. It is also concentrated: farmers with a low permanent component barely gain (Appendix Table 18). Full and efficient insurance does not make intrinsically unproductive farms—poor land, tiny plots, remote locations—productive. The productivity side is, moreover, conservative. I hold relative crop prices fixed—their adjustment could push the gains either way, and in any case most agricultural output in Uganda is not marketed. I hold the rural wage fixed, abstracting from cross-sector labor movement, and I abstract from land reallocation across farmers—both forces that would raise the gains and drive structural transformation. I abstract from these because they are hard to justify in an agrarian economy with incomplete land markets and subsistence-based production.

Yet, set against the high consumption insurance described in rural developing economies, the gains are large. Rural economies like Uganda appear, under the standard test that treats income as exogenous, close to fully insured, with a Townsend coefficient only about 0.04.[26] That near-complete smoothing, however, is achieved at a cost—the same incomplete markets distort households' income and investment choices enough to forgo the equivalent of 18% of consumption and welfare in this calibration.

Table 7 re-calibrates the model under alternative assumptions and reports its fit and the gains from completing markets.[27] The first alternative matches the low intermediate-expenditure share (0.10 in the data against 0.34 in the baseline), re-estimating the input elasticities α and γ; they fall to 0.11 and 0.14, well below the values estimated for agricultural production functions. This mechanically attributes the low input use to technology rather than to the friction, and the productivity gain collapses to 1.8%. The second drops bananas, addressing the concern that a single perennial crop drives the results. It does not: the risk trade-off between high- and low-yield crops increases, and the gains are not too different—10.3% in productivity, 12.0% in consumption, and 12.3% in welfare. The third recomputes the risk moments at the district level. Measuring risk from district-average yields removes household-level variation within districts—reducing measurement error but also removing household idiosyncratic risk. Total risk falls by roughly half, and the gains decrease but stay sizeable—5.6% in productivity, 7.1% in consumption, and 7.6% in welfare.

I also vary the depreciation rate $\delta \in [0, 0.15]$, the parameter that governs the cost of saving and hence the strength of the precautionary-saving channel. δ is calibrated to average liquid wealth; at the calibrated $\delta \approx 0.11$ the model also reproduces consumption volatility, which is

[26] Under CRRA preferences, the Townsend coefficient is the pass-through of idiosyncratic income into consumption, β, in $\Delta \ln c_{it} = \alpha \Delta \ln C_{rt} + \beta \Delta \ln y_{it} + u_{it}$, where aggregate regional consumption C_{rt} absorbs common shocks and I control for family size; $\beta = 0$ under full insurance. I estimate $\hat{\beta} = 0.0388$ (95% confidence interval $[0.024, 0.054]$).

[27] Each alternative re-fits the model—the scale parameters A and B, the shock variances, the permanent-productivity dispersion, and the depreciation rate δ—to match the crop yields, their risk moments, agricultural inequality, and average wealth, changing only the assumption noted.

World Development 207 (2026) 107497

Contents lists available at ScienceDirect

World Development

journal homepage: www.elsevier.com/locate/worlddev

Regular Research Article

Mobile money and the social contract: Experimental evidence from Ghana

Alex Yeandle [a,b,*], David Doyle [c]

[a] Nuffield College, University of Oxford, Oxford OX1 1NF, United Kingdom
[b] Department of Government, London School of Economics and Political Science, Houghton St, London, WC2A 2AE, United Kingdom
[c] Department of Politics and International Relations, University of Oxford, Manor Road Building, Manor Road, Oxford OX1 3UQ, United Kingdom

ARTICLE INFO

Keywords:
Mobile money
Tax morale
Public opinion
Africa
Experiment

ABSTRACT

Mobile money platforms are ubiquitous in many emerging economies. Hailed for raising financial inclusion and economic wellbeing, governments have now turned to mobile payments as a source of tax revenue. Transaction levies are often regressive, unpopular, and can encourage a return to cash. We present experimental evidence that they may also harm tax morale, a core component of the social contract between citizen and state. We present results from a survey experiment in Ghana, in which priming a controversial mobile transaction levy significantly lowers support for the state's right to collect taxes and willingness to comply with tax laws. We combine this with focus group discussions and analyse effect heterogeneity to examine two pre-registered explanations: transaction taxes cost citizens more than they expect to gain (reciprocity) and provoke particular backlash from non-government voters (partisanship). Our findings suggest that taxing mobile money can undermine efforts to expand fiscal capacity, while raising important mechanistic and policy questions for future research.

1. Introduction

In the wake of the Covid-19 pandemic, withdrawal of foreign aid, and shocks to global oil prices, developing fiscal capacity is increasingly important for emerging economies. But a range of structural constraints make this a difficult task to achieve. Stymied by large informal labor markets, powerful entrenched business interests, a lack of encompassing tax bureaucracies, and weak tax morale, the mobilization of much-needed tax revenue remains a serious challenge (Bastiaens and Rudra, 2018; Fairfield, 2015; Jensen and Weigel, 2026).

Unable to pursue conventional paths to taxation, policymakers in low-income states have turned to a range of digital technologies (Gupta et al., 2017; Ndulu et al., 2023; Okunogbe and Santoro, 2023). Often with the assistance of donors, this includes developing tools to biometrically identify citizens (Muralidharan, Niehaus, and Sukhtankar, 2016; Gelb and Metz, 2018; Garbe et al., 2024), digitize administrative records (Okunogbe, 2021; Casaburi and Troiano, 2016), and track the "information trails" of business activity in real time (Pomeranz, 2015; Mascagni, Mengistu, and Woldeyes, 2021; Lees et al., 2025). Taken together, these measures aim to strengthen fiscal capacity and thus provide the state with revenues to fund public spending.

Digitalization also opens up new modes of taxation to policymakers. Governments from several lower-income states, particularly in Sub-Saharan Africa, have begun placing levies on payments made through mobile money platforms. These platforms, which have digitalized vast swathes of previously informal economic activity, have become ubiquitous across the African continent. Citizens can now send and receive payments, store savings, run businesses, and access state support without the need for a conventional bank account (Aker and Mbiti, 2010; Mbiti and Blumenstock, 2015; Aron, 2018; Suri et al., 2023). As of 2024, there were more than 1.6 billion registered mobile money accounts, more than 400 million of which have been created since 2020 (GSMA, 2023). Across Africa, more than $1 trillion were transacted on these platforms in 2024 alone (Raithatha and Storchi, 2025). The adoption of mobile money has significant welfare-enhancing effects, which indirectly add to tax revenues as economies grow (Apeti and Edoh, 2023; Wandaogo, Sawadogo, and Lastunen, 2022). But by taxing mobile transactions *directly*, governments risk penalizing the lesser-off and encouraging a return to cash, undermining some of these gains (Anyidoho et al., 2023a; Ndungu'u, 2019).

In this paper, we study the microfoundations through which mobile transaction levies can shape economic and institutional development: their effect on tax morale. Public perceptions about the state's right to collect taxes, compliance with tax rules, and beliefs about the collective

* Corresponding author.
E-mail addresses: alexander.yeandle@nuffield.ox.ac.uk (A. Yeandle), david.doyle@politics.ox.ac.uk (D. Doyle).

https://doi.org/10.1016/j.worlddev.2026.107497
Accepted 18 June 2026
Available online 24 June 2026
0305-750X/© 2026 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

gains of taxation are crucial demand-side determinants of long-run fiscal capacity (Levi, 1988; Tilly, 1993; Fjeldstad, 2004; Martin, 2023; Jensen and Weigel, 2026). But without a social contract between citizen and state, policymakers are often subject to development "traps", where low capacity and poor public good provision become mutually reinforcing (Nathan, 2019) and citizens seek to opt out (Castañeda, Doyle, and Schwartz, 2020). We provide evidence that taxing mobile money payments can exacerbate this problem, by further harming tax morale among the public.

Using the case of Ghana, we present results from a pre-registered survey experiment (N = 1,151) in which priming a controversial mobile transactions levy reduces tax morale. These charges have been introduced across the African continent in recent years, from Kenya to Zimbabwe to Cameroon (Clifford, 2020; Lees and Akol, 2021; Gilbert, 2023). Ghana's version, known locally as the "e-levy", collected 1 % of all payments above 100 Ghana Cedi per day.[1] Introduced in June 2022, the government claimed this was to raise funds needed due to Covid and the resulting economic downturn (Osae-Kwapong, 2022). The tax also replaced revenues from toll booths, which were shut down due to concerns about corruption (Anyidoho et al., 2023a).

From the moment of introduction, the e-levy has been controversial and sparked anger over its regressivity and skepticism that revenues would be spent effectively (Osae-Kwapong, 2022). Indeed, around 18 months after our experiment was fielded, the policy was abolished after the opposition won the 2024 general election (GhanaWeb, 2025). This controversy was echoed in our experiment. We find that exposing respondents to a few short questions about the policy has significant negative effects on tax morale. Treated individuals are less likely to agree that the government has the right to collect taxes from its citizens, and are less willing to sanction someone who refuses to pay taxes that they owe.

We suggest two channels through which this negative effect occurs, which we evidence with qualitative and quantitative data. In focus group discussions, participants complained about the e-levy's high financial cost relative to the low likelihood it would improve public services, a channel we term *reciprocity* (see Levi, 1988). In addition, non-government voters saw the policy as a means for the ruling party to enrich its supporters, while being indicative of wider incompetence, a channel we term *partisanship*. When examining heterogeneity in our experimental estimates, the results are consistent with these channels. Negative treatment effects are driven by respondents most financially exposed to the tax, and those who did not vote for the government in the preceding general election.

Our findings have several implications. First, they are important for the study of fiscal capacity in low-income states. While existing work on tax morale highlights the importance of reciprocity, legitimacy, and institutional trust (e.g. Fjeldstad, 2004; Berens and Gelepithis, 2019; Castañeda, Doyle, and Schwartz, 2020; Weigel, 2020), we apply these frameworks to a new and less-studied type of taxation that is growing increasingly prominent. Mobile money taxes have been described as a "golden goose" in low-capacity contexts (Whitehouse, 2021), since, despite their ostensible ease for policymakers to collect, they risk pushing users back towards cash and thus shrinking the pot of available revenue. Our results help understand this dynamic, and, in so doing, contribute to wider debates about policy design and the importance of building public consent (e.g. Flores-Macías, 2018; Yeandle, 2025a).

More broadly, our results also add to recent work on the promise and pitfalls of digital revenue instruments, and how they can fail to produce desired gains in fiscal capacity (Okunogbe and Santoro, 2023). As policymakers leverage a vast new range of digital tools, there has been increasing focus on implementation problems (Chalendard et al., 2023; Bachas, Jensen, and Gadenne, 2024), particularly the strategic responses of citizens and firms (Mascagni, Mengistu, and Woldeyes, 2021; Lees et al., 2025), both of which ultimately undermine revenue collection. We, in turn, connect these debates to the broader *political economy* constraints faced by policymakers, namely the importance of public opinion and electoral politics in implicitly governing the types of tax that states can or cannot collect (Fairfield, 2015; Ardanaz, Hallerberg, and Scartascini, 2020; Martin, 2023). In this vein, our findings also help to contextualize the widespread watering down and withdrawal of mobile transaction taxes beyond Ghana, especially in countries where taxation and reciprocity concerns remain salient in political debate. From Malawi to the DRC to Tanzania, such levies have been abandoned or amended in response to the widespread backlash they invoke (Yeandle, 2025a; CGTN Africa, 2023; Gilbert, 2023).

The paper proceeds as follows. In the next section, we discuss some of the literature on taxation and tax morale in low-capacity contexts. Section 3 outlines our pre-registered experimental design, while Section 4 presents the main results and explores the potential mechanisms underpinning them. The final section concludes.

2. Building fiscal capacity but lowering tax morale?

2.1. Structural constraints to raising tax

Raising tax revenue remains a significant challenge for low-capacity states (see Jensen and Weigel, 2026). Much past literature has focused on the *structural* impediments to tax reform and domestic revenue mobilization. This includes the entrenched power of elite interests in preventing governments mobilizing tax revenue directly from corporate groups or wealthy sectors (Schneider, 2004; Fairfield, 2015), large informal labor markets that make much economic activity illegible to the state and thus hard to directly tax (Scott, 1998; Lee and Zhang, 2017; Castaneda and Doyle, 2019), different traditions associated with the rule of law in driving compliance (Kasara, 2007; Bergman, 2009; Sánchez de la Sierra, 2020; Weigel et al., 2021), and a general lack of bureaucratic capacity, particularly in more peripheral regions (Herbst, 2014).

Collectively, these structural impediments to taxation leave low-capacity states with low revenues and often regressive tax structures (Bachas, Jensen, and Gadenne, 2024). This makes it difficult for governments to provide high-quality and encompassing public goods, undermining tax morale and compliance among the public (Jensen and Weigel, 2026). Indeed, recent work has highlighted how the public do not believe that paying tax leads to discernible reciprocal benefits (Ali, Fjeldstad, and Sjursen, 2014; Bodea and LeBas, 2016a; Castañeda, Doyle, and Schwartz, 2020; Holland, 2018). This is exacerbated by low levels of trust in the ability of state to act (Berens and von Schiller, 2017; Cheeseman and Peiffer, 2023; Yeandle Forthcoming). Such low levels of tax morale and distrust in state institutions, together with easy-to-tax natural resource endowments, remove any electoral incentive to engage in meaningful tax reforms (see Ross, 1999; Humphreys, Sachs, and Stiglitz, 2007; Fairfield, 2015; Besley and Persson, 2014).

In light of these constraints, many low-income countries are increasingly reliant on consumption-based, rather than direct, taxation (Hart, 2010; Kato, 2003). These levies are often highly efficient and cost effective to implement, and indeed, across Sub-Saharan Africa, indirect taxes make up a larger share of revenues than in any other region (Bachas, Jensen, and Gadenne, 2024). However, this also carries some regressive implications, and there is some evidence that regressive tax structures can also negatively shape tax morale among citizens (Hoy, 2025).

2.2. Mobile transaction levies

Recently, a new potential source of revenue has emerged for governments in low-capacity contexts; mobile money platforms. Amid a broader process of digitalization (Okunogbe and Santoro, 2023), many

[1] Approximately 8.7 USD in September 2023, with exemptions for payments made to tax-registered businesses, government bodies, or transfers between one's own accounts.

emerging economies have rapidly moved away from physical cash and toward mobile-based payment platforms. These bring vast swathes of informal economic activity into the domain of the state, and therefore offer new routes for governments to expand the tax base. As can be seen in Fig. 1, uptake has been particularly rapid in Sub-Saharan Africa, where more than a trillion USD are now transacted every year via mobile payments (Raithatha and Storchi, 2025). This, combined with the world's lowest levels of fiscal capacity, have seen countries across the continent introduce various mobile transaction levies which collect a set percentage of each payment. While we lack systematic data on the adoption of these policies, a non-exhaustive list of countries that have attempted to adopt them, with varying degrees of success, includes Ghana, Gabon, Kenya, Côte d'Ivoire, Cameroon, the Democratic Republic of the Congo, Congo-Brazzaville, Malawi, Uganda, Zimbabwe, and Zambia (Clifford, 2020; Yeandle, 2025a).

A significant body of research, from both industry and the academy, has studied the *welfare* enhancing effects of access to mobile money services and their tax-raising potential (e.g. Aker and Mbiti, 2010; Jack and Suri, 2014; Aron, 2018; GSMA, 2023). For instance, there is some evidence that mobile money adoption can indirectly increase tax revenue, simply by making the economy grow (Apeti and Edoh, 2023). However, mobile transaction taxes may also have regressive implications. As a flat-tax they disproportionately affect the lesser-off, while tending to be paid by those more reliant on mobile money, namely informal workers rather than those in formalized sectors with access to a conventional bank account. Taken together, there is some fear that these taxes can incentivize a return to cash as lower-income citizens seek to avoid charges (Anyidoho et al., 2023a; Ndungu'u, 2019).

Despite this, we still know little about the precise ways in which these levies shape public opinion. In particular, these taxes might plausibly shape tax *morale*: public perceptions about the wider legitimacy of tax, beliefs about how revenues are spent, and willingness to pay. This is a significant gap, since tax morale is a crucial "demand-side" determinant of individual and collective compliance (Luttmer and Singhal, 2014; Jensen and Weigel, 2026). Over the longer term, it also underpins macro-level trends in fiscal capacity and state development (Levi, 1988; Tilly, 1993).

2.3. Tax morale

Existing work in low-capacity contexts suggests that transaction taxes, in general, should have negative implications for tax morale. We extend these arguments to a new and increasingly prevalent mode of digital taxation. We suggest two reasons for this negativity; mobile money taxes are generally not perceived as reciprocal among citizens and are prone to becoming split along partisan lines.

First, mobile money taxes are not *reciprocal*. Foundational arguments on the development of fiscal capacity stress that taxation is part of a broader bargain, or social contract, from which the public need to benefit (Levi, 1988; Tilly, 1993). This usually arises from the state's ability to leverage economies of scale and thus provide cheaper services than individuals can afford privately (Meltzer and Richard, 1981). In many low-income settings, however, public service provision is of dubious quality, does not reach the poorest citizens (Holland, 2018), or is less efficient than private substitutes (Castañeda, Doyle, and Schwartz, 2020).[2] Mobile money taxes are an acute example of an unreciprocated tax, falling disproportionately on poorer citizens most likely to rely on mobile platforms, but who are least likely to experience the fruits of government spending (Hoy, 2025). As a result, there is a high risk that these taxes are felt to be unfair by citizens, and an instance of the state not holding up its side of the bargain (Prichard et al., 2019; Castañeda, 2024).

Second, new taxes risk being seen through a *partisan* lens. On the one hand, governments around the world frequently direct resources to their own supporters (Stokes et al., 2013). In low-income settings this is often highly visible, due to inequalities in basic infrastructure (e.g. roads, schools, hospitals) that can arise over time and in response to electoral pressures (e.g. Burgess et al., 2015; Harding, 2020). This leads members of "out" groups to see taxation as less reciprocal than "in" groups, who have more to gain from the social contract when state services can be made excludable (Horowitz, 2019; Nathan, 2019; Green, 2020).[3] On the other hand, new taxes can signal government competence, perceptions of which may be shaped by pre-existing biases among voters. While work on partisan motivated reasoning is historically drawn from the American case (Campbell et al., 1980; Flynn, Nyhan, and Reifler, 2017), there is growing evidence of similar dynamics in African democracies (e.g. Carlson, 2016; Harding and Michelitch, 2019; Mattes and Krönke, 2020), including Ghana (Yeandle and Doyle, 2025). As a result, government supporters expect to gain more from tax revenues, and are more supportive of government choices, so will view mobile money levies less negatively.

2.4. Empirical expectations

We apply these frameworks to the case of Ghana's electronic transaction levy ("e-levy"), a controversial mobile money tax introduced in June 2022. In a survey experiment, we randomly expose respondents to questions designed to prime the e-levy and perceptions of it. We anticipate that our prime will trigger a negative response from treated participants, given the policy's deep unpopularity. This, we expect, will drive changes in how they respond to questions about taxation.

We focus on three related outcomes; legitimacy, compliance, and redistribution preferences. The first two directly relate to tax morale. If respondents view authorities as *legitimate*, it suggests that they hold the normative belief that collecting taxes is an appropriate use of state power (Luttmer and Singhal, 2014; Dwenger et al., 2016). *Compliance* taps into whether citizens feel obligated to follow existing tax rules. In contexts of high state capacity, the government can (coercively) enforce these rules, and so legitimacy is not a necessary precondition for compliance. However in lower-income countries, where states are weaker, public consent matters much more in driving so-called "quasi-voluntary" tax payments, for instance by choosing to make oneself legible to the government by reporting income and assets (Levi, 1988; Khan et al., 2022; Grieco et al., 2025). These two measures aim to capture both the belief system that underpins tax morale (legitimacy) and a behavioural outcome that follows on from it (compliance). In general, we expect that these measures will move in tandem; higher perceptions of legitimacy will correlate with higher compliance, and vice versa (Jensen and Weigel, 2026). However, as we discuss in the mechanisms section, there are some reasons why they may diverge for certain groups.

In addition to these, we also study generic preferences for *redistribution*. While not a core component of tax morale, this is instead designed to capture general preferences for expanding the size of the state through taxation. Redistribution preferences are most commonly tied to where a respondent situates themselves on the income distribution, and therefore whether they feel likely to win or lose out from taxation (Meltzer and Richard, 1981). To the extent that the e-levy undermines respondents' perceptions that the government will equitably invest tax revenues into public goods, we might expect support for redistribution to fall (Holland, 2018). Alternatively, respondents might maintain a preference for redistribution *in principle*, while viewing the e-

[2] There is also evidence to support the *reverse* argument: where public services are better than expected, this can foster norms toward tax compliance (e.g. Bodea and LeBas, 2016b).

[3] See Kasara (2007) for an alternative account, whereby the state is sometimes better placed to *enforce* taxation rules among its supporters via social institutions.

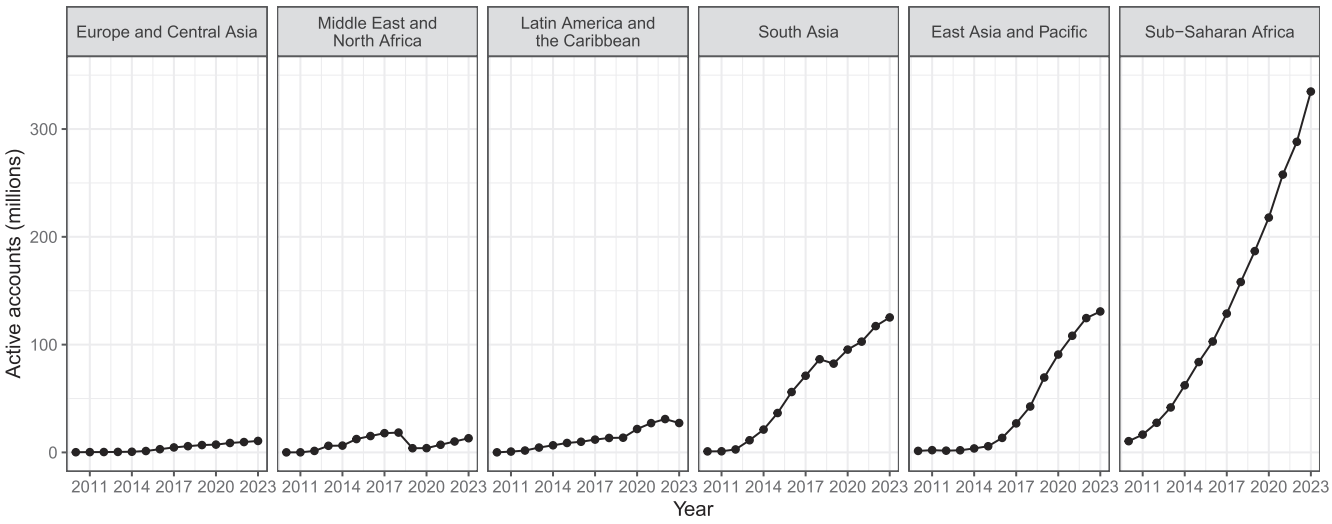


Fig. 1. Growth of mobile money accounts. Source: GSMA (via Our World in Data).

levy as a poor mechanism by which to achieve it. In this case, we would only see movement on those outcomes more directly connected to tax morale.[4] As a baseline therefore, we pre-registered the following hypotheses:

H1 (Tax legitimacy): *Those in the treatment group will be less likely to agree that tax authorities have the right to make people pay taxes, compared with those in the control.*
H2 (Tax compliance): *Those in the treatment group will be more likely to agree that people should refuse to pay taxes, compared with those in the control.*
H3 (Redistribution): *Those in the treatment group will be less supportive of tax-based government redistribution, compared to those in the control.*

3. Research design

3.1. Data

Our experiment was embedded within a broader survey conducted in Ghana in September/October 2023. The survey studies various ways in which financial remittances, mobile phone access, and exposure to bribes shape public opinion in Ghana. To this end, we strategically oversampled respondents who receive remittances, with around 59 % receiving funds from relatives over the previous 12 months.

The survey took place in three regions across the country: Greater Accra, Central, and Bono. Fieldwork was implemented by the Ghana Centre for Democratic Development (CDD), a local research firm who also administer the Afrobarometer survey in Ghana. We selected respondents through a multistage sampling process in which communities were chosen at random using the most recent round of the census, and individual respondents within communities were selected using a random-walk protocol similar to that used by Afrobarometer. Enumerators were required to meet a minimum quota of remittance recipients in each community, and, where possible, to alternate interviews by gender. Our final survey yielded 1,151 respondents: 450 from Bono, 451 from Greater Accra, and 250 from Central.

In section A of the supplementary materials we provide further detail on sampling strategy and survey fieldwork. We also formally compare respondents with those from a recent, nationally representative, round of the Afrobarometer. While we do not claim that our sample is nationally representative, it is largely similar, demographically and politically, to population as a whole. Moreover, given the nature of our experimental design and particular impact of the e-levy on remittance payments sent through mobile money, our quota approach provides greater leverage to analyze heterogeneous effects that a nationally representative survey would not provide.

3.2. Intervention

Our experimental intervention was simple. Respondents randomly assigned to the treatment group were asked three questions about Ghana's e-levy. For those in the control, these questions were skipped and respondents moved straight to the outcome measures we use to capture tax morale. By asking these questions, we aimed to prime respondents' initial response to the policy and to highlight its salience when responding to our measures of tax morale downstream. Fig. 2 below, reproduced from our pre-analysis plan, summarizes this process.[5] Additional detail can be found in section A.2 of the supplementary materials.

In Fig. 3, as well as section A.4 of the supplementary materials, we present balance tests to show that respondents' political and demographic characteristics do not predict treatment assignment. Collectively, the model also provides a poor predictive fit (F stat = 0.74, p = 0.72). This gives confidence that randomization was successful, and that our treatment and control groups are therefore similar in all respects but the e-levy prime.

Those in the treatment group were asked the following three questions:

Treatment question 1: *In June 2022, the government introduced the electronic transaction levy, or e-levy, which charges a 1 % tax on electronic money transfers above 100GHS. The government can use e-levy revenues to pay for public development projects and services. Were you aware of the e-levy before this interview? [Yes, No, Don't know]*
Treatment question 2: *Have you personally paid the e-levy charge in the past twelve months? [Yes many times, Yes a few times, Yes once or twice, No not at all, Don't know]*

[4] We also pre-registered additional hypotheses about perceptions of government corruption. These were not part of our core measures of tax morale, but are reported in full in section D2 of the supplementary materials. Contrary to expectations, we find null effects on all measures of corruption. This is likely because citizens have high, and strong, prior beliefs about corruption.

[5] Our pre-analysis plan can be found at osf.io/s9utc.

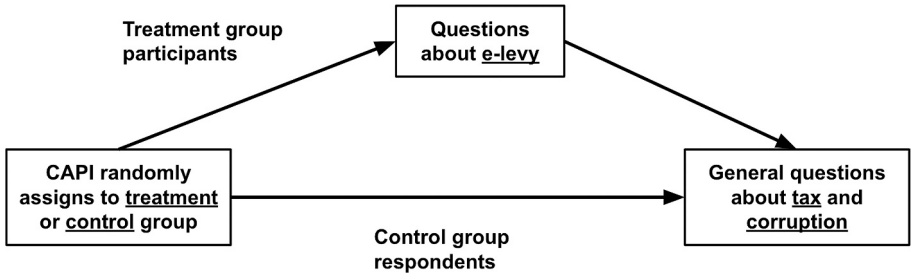


Fig. 2. Flow diagram of the experimental intervention.

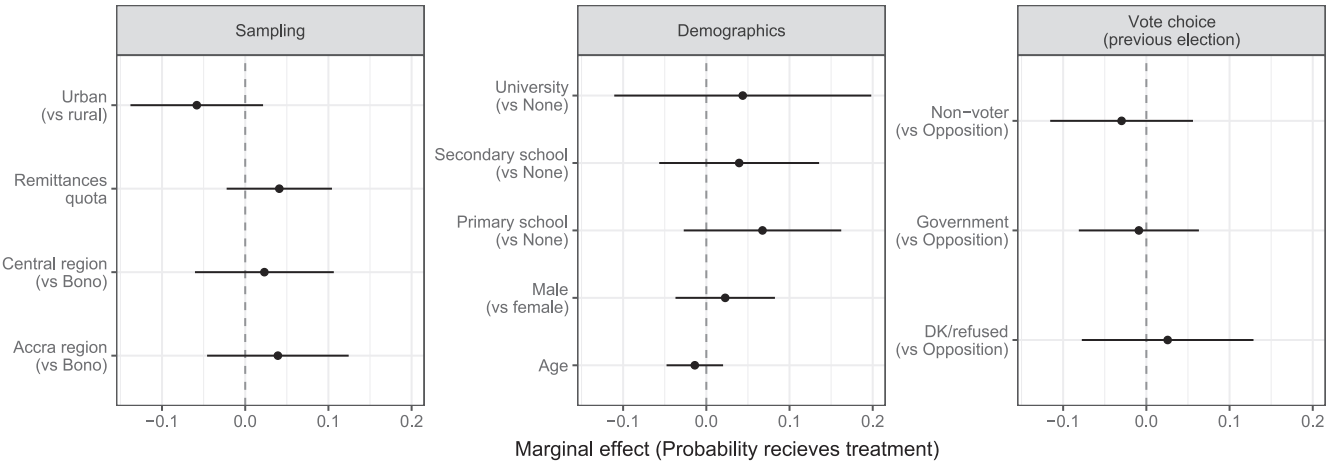


Fig. 3. Balance checks.

Treatment question 3: *Do you think ordinary Ghanaians have benefited from the money raised by the e-levy? [Yes they have benefited, No they have not benefited, Don't know]*

Fig. 4 plots how treatment group participants responded to these questions. Consistent with our expectations, there is high awareness of the e-levy among our sample (93 %), broad consensus that the policy has *not* benefited ordinary people (92 %), and evidence that the most respondents (67 %) have paid the charge at least once in the last year. These findings a similar to those from a nationally representative Afrobarometer survey fielded a few months prior, in which 75 % of respondents disapprove of the levy, 76 % believe it will increase financial burdens on the poor, and 49 % intend to use cash payments to avoid the charge in future (Osae-Kwapong, 2022). Collectively, these results suggest that our treatment is *priming* respondents to think about an issue already known to them, rather than *informing* them of something new (Dunning et al., 2019).

3.3. Outcomes

Prompting our treatment group with these questions allows us to measure the effects of priming the e-levy on respondents' tax attitudes and preferences. We focus on three measures: legitimacy of tax authorities, compliance with tax rules, and generic preferences for redistribution. Respondents were asked about the extent to which they agree or disagree with the following statements about taxation in Ghana, each measured on a 4 point scale. For question (b) we reverse the response scale, so that more positive outcomes indicate greater compliance.

a) [Legitimacy] "Tax authorities always have the right to make people pay taxes to the government."
b) [Compliance] "Sometimes people should refuse to pay taxes that they owe to the government."
c) [Redistribution] "The government should reduce income differences between the rich and the poor, by taxing wealthy families and giving assistance to the poor."

Fig. 5 presents the baseline distribution of each question among control group respondents. There are relatively high baselines across each outcome: most control group supporters do view the state as legitimate (mean 3.1/4); do not think people should avoid taxes (mean 2.9/4), and support the general principle of economic redistribution from rich to poor (mean 2.9/4).

3.4. Estimation

Following our PAP, we then use a variety of specifications to estimate treatment effects. At baseline, for respondents i living in region r, we measure the effect of treatment on tax outcomes y. We then vary whether we include a) a vector of individual-level demographic covariates X (education, age, gender and urban/rural), and b) these same demographic covariates in addition to a measure of how the respondent voted in the previous election. On top of these individual-level characteristics, we also vary the inclusion of region fixed effects α_r. Further discussion of these variables, and of any deviations from our pre-analysis plan, can be found in sections A and B in the supplementary materials.

In total this gives 6 specifications for each outcome, all of which we present in the results section: baseline, demographic, and demographic + political characteristics, each with and without region fixed effects. While our randomized treatment means that results should not diverge between specifications, our inclusion of additional variables might help to increase the precision of our estimates (Gerber and Green, 2012) and reinforces confidence that our findings are not driven by residual differences between groups.

$$y_{\{ir\}} = \beta_1 (treated)_{\{ir\}} + \gamma X_{\{ir\}} + \alpha_r + \varepsilon_{\{ir\}}$$

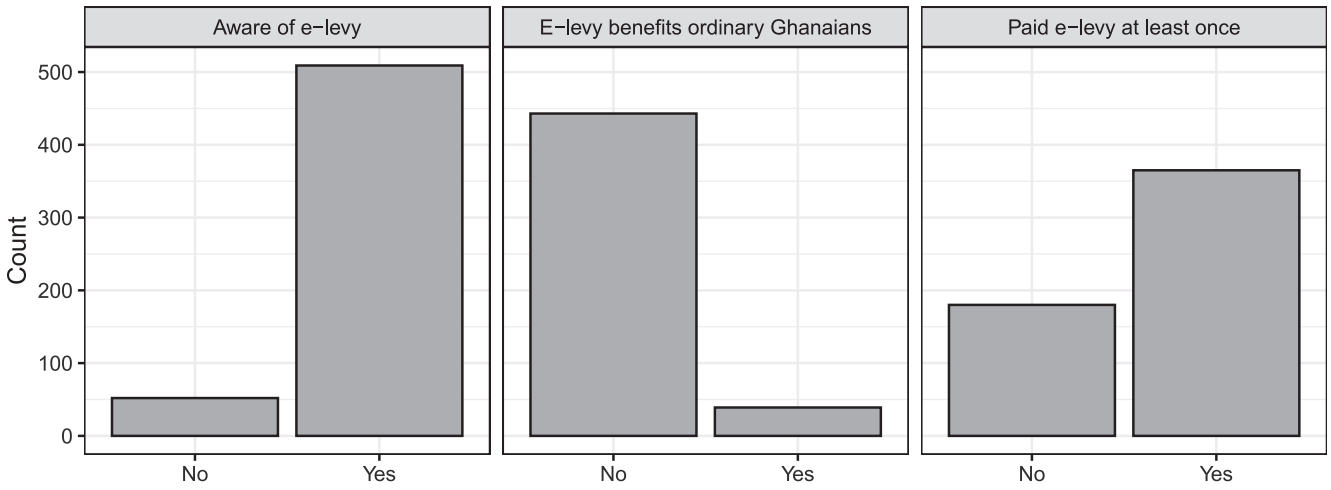


Fig. 4. Perceptions of the e-levy, excluding don't knows. Questions only asked to respondents randomly assigned to the treatment group (N = 563).

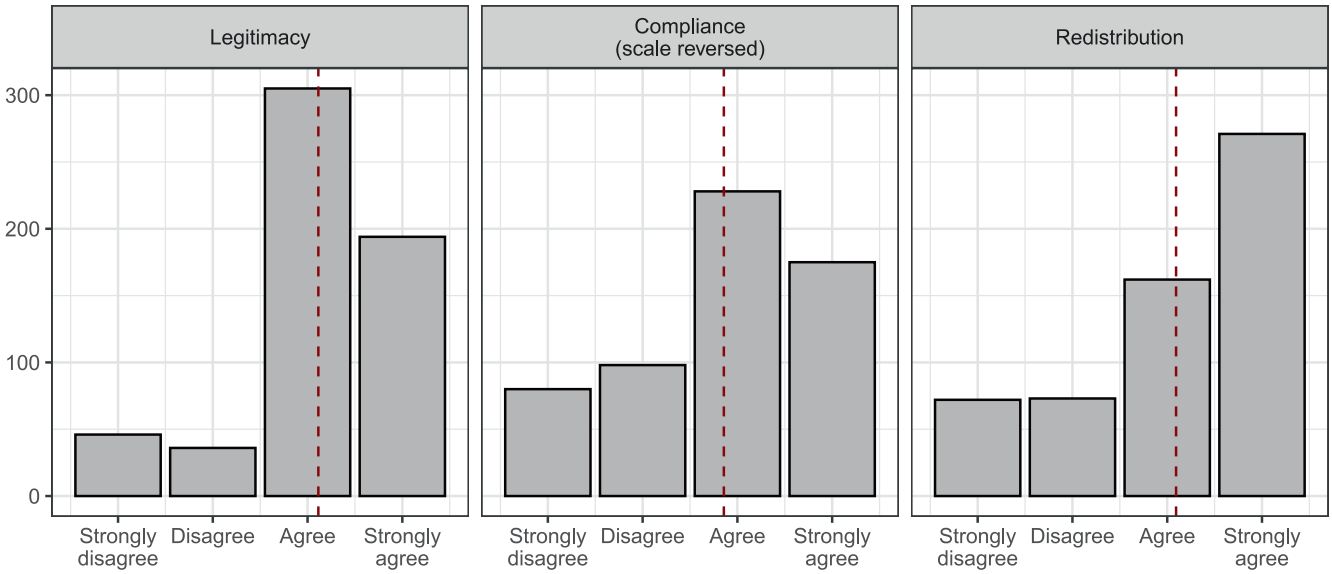


Fig. 5. Distribution of tax outcome questions (control group respondents only).

4. Results

4.1. Main findings

We present our main results in Fig. 6, with full regression tables in section D of the supplementary materials. The left-hand pane reports the average treatment effect for our question capturing legitimacy; the middle pane represents the average treatment effect for compliance; and the right-hand pane represents the effect of the treatment on the third question capturing preferences to use taxation to reduce the gap between the rich and the poor. The gray point estimate is the raw model, with the treatment only; the orange point estimate includes demographic controls; and the blue point estimate includes vote choice in the previous election. Each pane depicts models without (left), and with (right), region fixed-effects. The bars represent 90 % and 95 % confidence intervals.

There are highly consistent effects on our two measures of tax morale (legitimacy and compliance). Priming the e-levy reduces the perceived legitimacy of the state to collect taxes, while also reducing respondents' compliance. In both cases, the effect size is around 0.1 on the 4-point agree-disagree response scale, equivalent to around 10 % of a standard deviation. These effects are consistent across each of the 6 specifications, and are significant at the 90 % or 95 % level throughout.

By contrast, the treatment has no effect on our redistribution outcome. While we had anticipated that priming an unpopular tax may reduce respondents' generic preferences for the state to redistribute resources, this does not appear to be the case. One explanation for the lack of effect here might be that our tax morale outcomes are somewhat *process* oriented, concerning how the state collects tax and respondents' views about whether this is legitimate and something with which to comply. Our measure of redistribution preferences, by contrast, is more *outcome* oriented, since it is focused on the state's larger role in moving resources from rich to poor. It might be that respondents have well-formed, stable preferences about this type of redistribution as a point of principle. In turn, while respondents might view the e-levy as a bad way to bring about this wider goal, wider preferences for the goal still remain unchanged.

More broadly, though, our main findings indicate that tax morale (legitimacy and compliance) decline significantly when the e-levy is made salient. This is despite a deliberately light-touch experimental treatment, which primes respondents to think about a policy with which they are familiar rather than providing them with new information. On the one hand, this suggests that our prime mimics a much deeper process

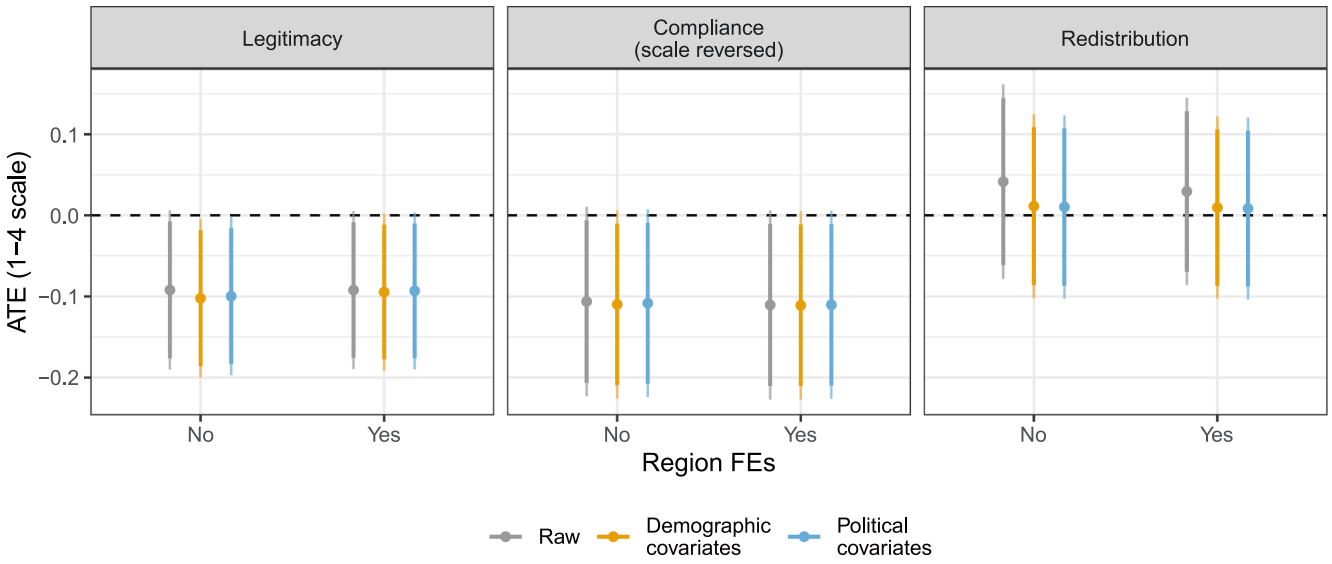


Fig. 6. Main results of the experiment. Estimates represent average treatment effects, including pre-registered demographic or political covariates, with and without region fixed effects. Bars represent 95 % (thick) or 90 % (thin) confidence intervals. Results suggest that priming the e-levy reduces perceptions of tax legitimacy and compliance, with no effect on redistribution. Full results in section D1 of the supplementary materials.

of attitudinal change brought about when the policy was first introduced, and voters learned about it for the first time. On the other, it suggests that real-life events that prime it, for instance discussions in the media and in election campaigns, can also have tangible effects on tax morale among the public.

4.2. Mechanisms

What underpins our main results? We examine two pre-registered explanations for why priming the e-levy would reduce reported tax morale: reciprocity and partisanship. We adopt a series of treatment-by-covariate specifications to examine heterogeneous effects, together with data from a series of pre-treatment focus group discussions, held in four regions across the country, to understand public backlash to the policy.[6] It is important to note that while these analyses can provide suggestive support for mechanisms, they do not yield the same direct, straightforward, causal interpretation of our main results (Gerber and Green, 2012). Nonetheless, we see these findings as a useful contribution that can help explain and contextualize our experimental estimates and help to motivate future empirical and theoretical work.

4.2.1. Reciprocity

We begin with reciprocity. Based on the seminal contribution of Levi (1988), reciprocity is underpinned by the idea that citizens are more willing to pay taxes when they expect to receive something in return. If public services are of higher quality, citizens will be more willing to pay taxes and are more likely to trust the government to use tax revenues effectively for the greater good. There is now ample evidence that attitudes towards taxation, particularly in low capacity contexts, are shaped by citizens' sense of reciprocal benefit and wider trust in government (Berens and Gelepithis, 2019; Berens and von Schiller, 2017; Castañeda, Doyle, and Schwartz, 2020; Holland, 2018).

Priming the e-levy appeared to trigger this sense of lacking reciprocal benefit among our focus group participants. The tax introduced financial costs for the public, but these did not correspond to visible improvements in public services. Participants complained that "*we are paying money and we don't get anything for it at all*", that "*[the government] are not using [e-levy revenues] to make the country better*", and that "*we don't get development…if you see some development it would be okay*".[7] There was also anger that revenues from the tax did not "*go into the right things. If there are roads and schools we should see it by now.*"[8]

This was linked to perceptions that the e-levy was inherently unfair, given mobile transactions are already taxed in other ways. For instance, in Ghana mobile money users pay charges when withdrawing cash from agents, some of which goes to the state. This was a sore point, with one respondent noting that "*my salary is being taxed so much. One money you pay tax 5 or 6 times. The work does not pay for these extras… I get less*".[9] There was further angst at taxes being collected from remittances. One sender, for instance, noted that "*I send money for the children in school but I still have to pay [the e-levy].*"[10]

This qualitative evidence from focus groups is reinforced when we examine effect heterogeneity in our experimental data. Following our pre-analysis plan, we explored whether treatment effects come from those respondents with the greatest financial exposure to the e-levy. Our results indicate that negative effects are largely driven by remittance recipients, mobile money users, and those who believe the government has little to no impact on their economic conditions.[11] These results are very much consistent with the reciprocity channel; these groups of respondents are the ones most directly exposed to the e-levy, and who plausibly have little to gain.

Firstly, we test whether treatment effects are more negative among those receiving financial remittances. As noted, our survey deliberately over-samples remittance recipients and so has a reasonable number of respondents in each group. In Ghana, remittances are very often sent or received by mobile money; indeed, in our survey 92 % of recipients report sometimes or always receiving funds in this way. This means that remittances are a strong proxy for potential exposure to the e-levy. We

[6] We conducted ten focus groups across Greater Accra, Central, Northern and Savannah. Each discussion took place with 6–10 participants, who were selected in conjunction with a local contact, knowledgeable of the community. Two discussions were all male, two all female, and the remainder were gender balanced. See section E of the supplementary materials for more details.

[7] Greater Accra, Aug 2023.
[8] Tamale, Sep 2023.
[9] Central region, Sep 2023.
[10] Tamale, Sep 2023.
[11] These were all pre-registered, with the exception of mobile money users which we discuss in supplementary materials section B.

measure remittance receipt among treatment and control group respondents, enabling us to run a conventional treatment-by-covariate interaction.

Fig. 7 depicts the effects across a binary measure of remittance use, coded as 0 for those who never receive remittances (N = 462, 40 %), and 1 for those who at least sometimes receive these payments (N = 684, 60 %). Full results tables can be found in supplementary materials section D.3. The results show that declining legitimacy and compliance are generally driven by those receiving remittances, while respondents not in receipt exhibit relatively precise null effects. These results are particularly pronounced for compliance, where the difference between subgroups is itself statistically significant (Gelman and Stern, 2006). Lower levels of compliance might reflect the fact that remittance recipients in Ghana are more likely to pay bribes to public officials, and so are more likely to have experience personally evading tax (Yeandle and Doyle, 2026).

Second, in section D.3 of the supplementary materials, we explore how effects differ between users and non-users of mobile money. To do this, we use a pre-treatment question on how often respondents use mobile money platforms, and separate respondents who never use these platforms (N = 244) from those who do (N = 906). Again, this question proxies personal exposure to the e-levy, with the implication being that mobile money users are more likely to bear direct financial costs from the policy and so be more hostile to it. This is broadly reflected in the results, with mobile money users driving the overall negative treatment effects we observe. These effects are particularly pronounced for compliance.

We also examine whether treatment effects differ across levels of attribution. We ask respondents the extent to which they think the government has influence over their living conditions, and code high attribution individuals as those who answer "Somewhat" or "A lot" (N = 872), and low attribution as answering "Not at all" or "A little" (N = 278). Our expectation is that lower attributing individuals will respond more negatively to the e-levy. In keeping with a reciprocity mechanism, these individuals will be less likely to believe the money they spend on taxes will come back to them through public services, so the e-levy represents a heavier, less reciprocated, financial loss. Our results, in section D.3 of the supplementary materials, offer some evidence that this is the case. Respondents with low attribution respond negatively to our e-levy treatment for our measure of legitimacy, but results are mixed for compliance (and continue to be null for redistribution). We suspect that part of the reason compliance estimates are noisy may stem from lower than expected variation in our attribution measure. We expected responses to this measure to be relatively evenly spread; however, baseline attribution was in fact very high (mean 3.1/4).

Finally, we also find that personally paying the e-levy has nuanced effects. Of our 563 treated respondents, 168 (30 %) have never paid, while 395 (70 %) have paid at least once over the past year. We split the treatment group into two groups, accordingly, and compare each to the control. The results, in supplementary materials section D.3, show that legitimacy falls among those who have paid the tax in the past twelve months, but it is those who have *not* paid that drive negative compliance effects. These findings diverge slightly from our pre-analysis plan, where we expected more negative effects from payers across the board. On reflection, however, these results do make some sense within a reciprocity framework. Those who pay the e-levy express frustration, by viewing the government as illegitimately taking their money. Those who do not pay have an interest in continuing to not pay, and so are less willing to sanction others who avoid paying the tax.

Taken together, these heterogeneous effects, paired with the focus group data, lend credence to the role of reciprocity in explaining the e-levy's negative effects on tax morale in Ghana.

4.2.2. *Partisanship*

In a second step, we explore the role of partisanship and the politicised nature of the e-levy. Our expectation is that government voters will be more receptive to efforts by the state to raise revenues, given the state is controlled by their preferred party and so is seen as more likely to benefit their interests. Moreover, because Ghana's main opposition pledged to abolish the e-levy, describing it "*distortionary and burdensome*", our treatment might prime partisan talking points about the policy and wider affective polarization (Akomah and Allotey, 2022; GhanaWeb, 2025).

In section C of the supplementary materials, we first show that baseline attitudes towards the e-levy and tax morale are shaped by party loyalty. Compared to opposition voters, government supporters are more likely to think the e-levy benefits ordinary Ghanaians, more likely to report paying it, more supportive of the state's right to collect tax, and more willing to comply with tax rules. This is consistent with a range of other studies showing that, despite widespread dislike of the policy

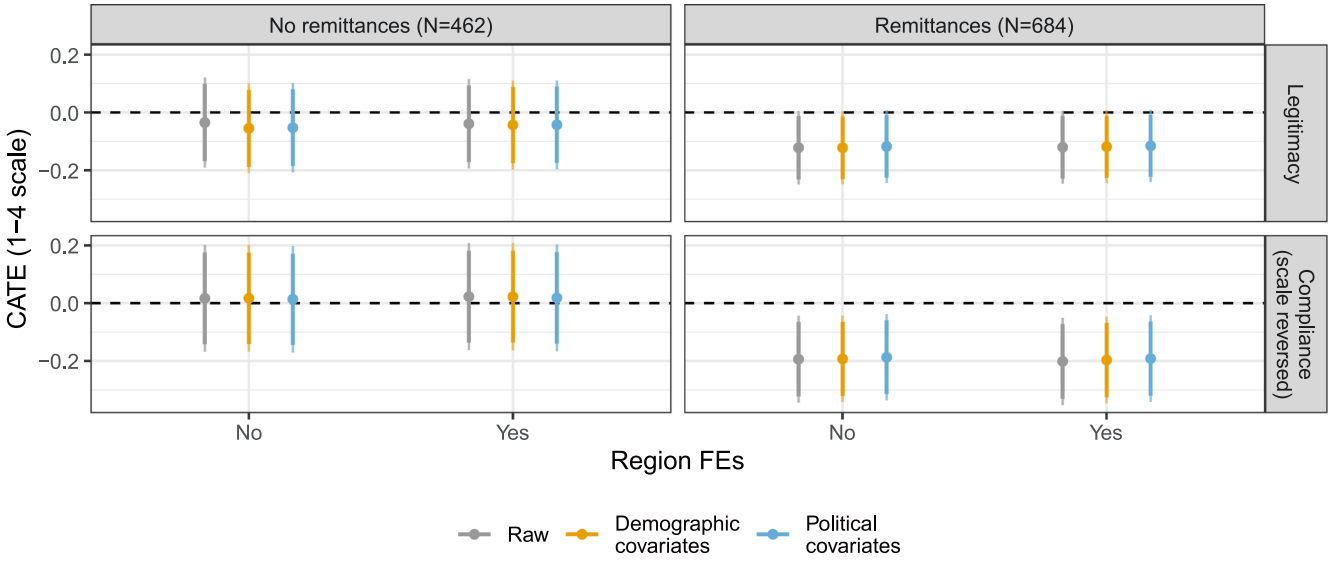


Fig. 7. Heterogeneous effects by remittances receipt. Estimates represent conditional average treatment effects for respondents who never vs at least sometimes receive remittances (measured pre-treatment). Bars represent 95 % (thick) or 90 % (thin) confidence intervals. Results suggest that treatment effects on legitimacy and compliance are driven by respondents receiving remittances. Since most remitted income is sent through mobile payment platforms, this group is likely to be more directly exposed to the tax. Full results in section D3 of the supplementary materials.

across Ghana (Anyidoho et al., 2023b; Kipo-Sunyehzi and Asante, 2025), the extent of this backlash varies by partisan preference (Dzordzormenyoh, 2025).

In focus group discussions, there was some evidence of these dynamics taking place. Many opposition supporters were not *a priori* opposed to the e-levy, but argued that the government had mishandled it. One described how "*I support [the e-levy] but they introduced it badly. The tax is okay, rich countries have taxes it's what we need. But [the government] implemented it badly*".[12] There was also evidence that opposition voters saw the policy as a way for the government to enrich itself and its backers. In the words of one, "*the NPP know they'll lose [the 2024 election] so they're trying to take what they can now*".[13] Many linked the tax to ongoing corruption allegations involving government figures, remarking, for instance, that "*the taxes should be going for something, now it's going to somebody's house*" – a reference to a recent discovery of millions of dollars at a minister's private home (Nunoo and Zane, 2023).[14]

Following our pre-analysis plan, we also examined whether our overall negative effects on tax morale vary by partisan group. We use pre-treatment measures of vote choice in the previous Presidential election, categorizing respondents as "Government" if they voted for the ruling New Patriotic Party (N = 397), "Opposition" if they voted for another party (N = 382), and "Don't know" if they could not recall their vote (N = 126).[15]

We present results in Fig. 8 with full models in section D4 of the supplementary materials. Consistent with a partisanship channel, government supporters do not see any significant negative change in tax morale upon receiving treatment. On compliance, we see a negative effect among opposition voters, while those who cannot recall their 2020 vote become more negative on legitimacy as well. One interpretation of these findings is that those with weak partisan ties (namely those who cannot recall their vote) are more influenced by government policies, and so the e-levy can have a significant impact on political attitudes. Another possibility is that government officials are better able to enforce tax rules with co-partisan voters, and perhaps better able to provide preferential treatment in return, which would explain a lack of compliance effect (Kasara, 2007). It is important to note, however, that we do not find a statistically significant difference in point estimates *across* each subgroup (Gelman and Stern, 2006). This is likely because of both the smaller sample sizes involved, and possible floor effects from opposition supporters having lower baseline levels of tax morale to start off with (as shown in supplementary materials section C).

Taken together, these analyses do suggest that partisan identities play a role in understanding perceptions of the e-levy. As ever, though, future work is required to fully disentangle these dynamics.

4.3. Extensions

We explored two additional tests with our data, both of which can be found in section D.5 of the supplementary materials.

First, as specified in our pre-analysis plan, we present additional results using a dichotomized outcome variable. Respondents who "agree" or "strongly agree" with a given statement are coded as 1, while those who "disagree" or "strongly disagree" as 0. In general, effects point in the expected direction but fail to reach statistical significance. There are two likely reasons for this. On the one hand, as we show in Fig. 5, there are high baseline levels of tax morale in our sample; the vast majority of respondents in the control group agree or agree strongly with each statement. This means that the binary outcome measure removes more variation than we had anticipated, making it harder to detect significant effects. Alternatively, this could reflect a qualitative difference in response scales; respondents were first asked if they agreed or disagreed with each statement, and then probed for the strength of that feeling (i.e. if they do so "strongly"). Our results plausibly suggest that the e-levy treatment shifts responses in the second step, reflecting a change in intensity without persuading respondents to change their mind. This makes sense amid the wider priming nature of our treatment and respondents' strongly held prior beliefs about the tax. Of course, though, we cannot fully distinguish which of these factors is more important, nor be wholly sure about which points in the ordinal scale are driving effects.

Second, we explore how the e-levy prime affects a range of other post-treatment outcomes. Specifically, we focus on trust in the President, trust in the police, trust in government officials and intention to vote for the ruling party. Priming the e-levy had no effect on trust in the President, police, or intention to vote for the government. This would appear to be in line with a partisanship channel, in which political allegiances are relatively fixed identities that shape responses to the treatment, rather than being shaped by it. We do, however, see that trust in "government officials" declines significantly. This is interesting, given that officials are less clearly partisan, and more akin to "the state" and wider system of government in general. They are also the central point of contact for accessing basic services, particularly for those receiving remittances in contexts where the private sector is weak (Yeandle and Doyle, 2026). In turn, then, attitudes toward officials are quite clearly connected to the taxes citizens pay and thus the reciprocity channel that we describe.

5. Discussion

As consumers across the Global South switch away from cash and toward mobile money, the nature of the social contract has undergone profound change. Mobile payment platforms make new types of economic activity legible to the state, and transaction taxes offer a quick and simple way for governments to extract revenue from areas previously beyond their reach. In this paper, we point to ways in which these short-run gains in revenue risk longer-term political costs: mobile money taxes weaken tax morale, are politically contentious, and risk reversing the positive welfare effects of mobile payments. We document this with evidence from a survey experiment in Ghana, where priming a new mobile transaction tax significantly undermines perceptions of tax legitimacy and increases support for evasion.

Drawing on general theories of tax morale and the social contract, we argue that mobile money levies are constrained by two political barriers. First, as a form of consumption tax, they are regressive and thus disproportionately affect those on the lowest incomes. This undermines citizens' *reciprocity* beliefs, since it is difficult for would-be payers to connect the tax to tangible benefits in the form of public goods and services. Second, building on this grievance, these taxes can quickly become points of *partisan* contention. In low and middle-income countries, such divides can reflect both performance evaluations about how competently the government implements the tax, but also reciprocity-oriented concerns if public spending is felt to disproportionately benefit government voters. They can also prime general party-political arguments, especially if opposition groups campaign against the tax and pledge to remove it.

Using Ghana's controversial "e-levy", a 1 % tax on mobile money transactions, we draw on an experiment embedded in an original survey to evidence the argument. Our simple intervention, where treated respondents are asked a handful of questions designed to impartially prime the policy, drives a significant reduction in perceptions of tax legitimacy and compliance. Our supplementary analyses, while not causal in their own right, include examining heterogeneous treatment effects and drawing on qualitative data from across Ghana. These point

[12] Central region, Sep 2023.
[13] Accra, Aug 2023.
[14] Central region, Sep 2023.
[15] We do not include estimates for those who did not participate (N=246), as the political leanings of this group is ambiguous and thus it is not clear how to expect them to behave.

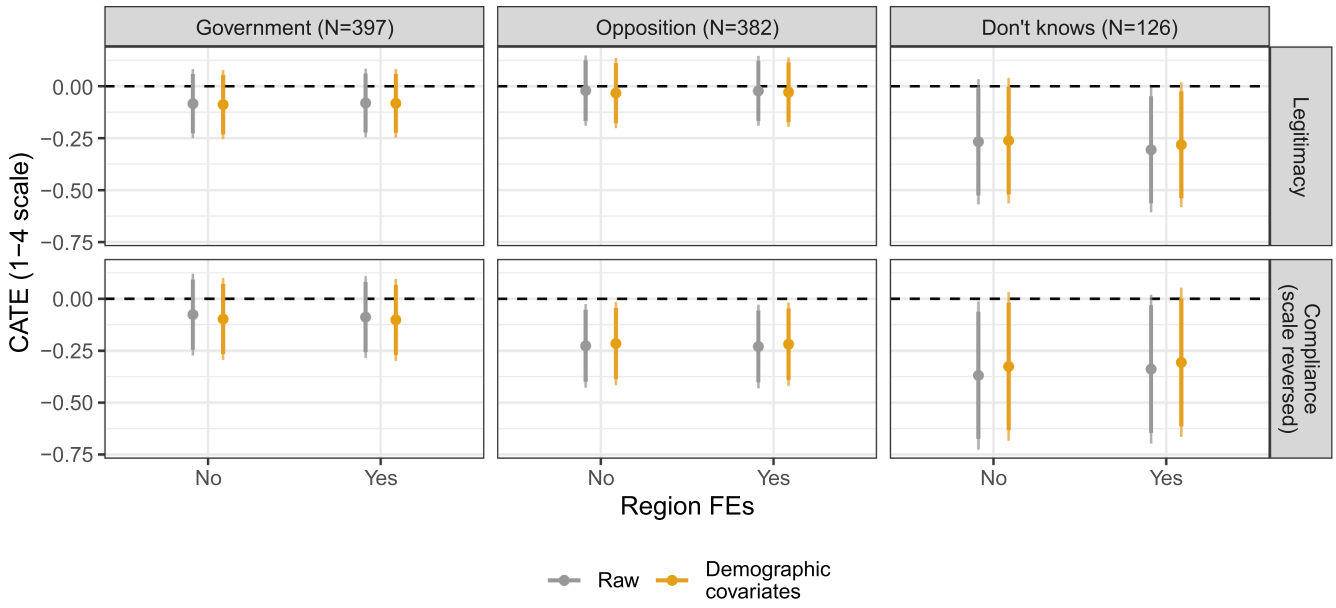


Fig. 8. Heterogeneous effects by 2020 vote choice. Estimates represent conditional average treatment effects for self-reported party choice in the previous general election (measured pre-treatment). The results suggest that government supporters do not respond to the e-levy prime for either outcome, while declining legitimacy is driven by don't knows and compliance driven by opposition and don't knows. Full results, including redistribution and corruption measures, are in section D4 of the supplementary materials.

to the importance of reciprocity beliefs, personal exposure to the tax, and partisan divides in explaining these effects.

Taken together, our findings make a series of contributions to the literature. Primarily, we add to a growing body of work on the political economy of taxation and the social contract in the Global South (Fairfield, 2015; Holland, 2018; Berens and Gelepithis, 2019; Martin, 2023). By concentrating on mobile transaction levies, we address a new and increasingly common policy lever used by resource-constrained governments, particularly in Sub-Saharan Africa (Clifford, 2020; Mader, Duvendack, and Macdonald, 2022). We show that these taxes face similar *political* constraints to their conventional counterparts, particularly around the visibility of benefits that citizens receive (Levi, 1988; Tilly, 1993) and the role of partisanship in forming such evaluations (Stokes et al., 2013; Carlson, 2016). Tax and reciprocity concerns are salient political issues in many African countries, and so our findings can also advance understanding of why mobile money taxes have been amended or withdrawn by politicians in response to public pressure across the continent (Mader, Duvendack, and Macdonald, 2022; Osae-Kwapong, 2022; Yeandle, 2025a). The case of Ghana also, perhaps more uniquely, exemplifies how these taxes can themselves become live issues in electoral campaigns (Dzordzormenyoh, 2025). This taps into wider arguments about why politicians may rationally choose to under-tax their citizens to strategically avoid raising expectations of public spending from voters (Martin, 2023).

Our focus on mobile money also adds to a growing literature on the wider political consequences of digitalization in low-income countries. Digitalization is often heralded as an economic transformation, with well-documented welfare-enhancing effects for citizens and the potential to make governments operate more efficiently (Gupta et al., 2017; Akitoby et al., 2020; Ndulu et al., 2023). Our findings instead echo a growing body of work that signals a more cautious approach. Rather than focusing on the technical and strategic difficulties of implementing new procedures (Mascagni, Mengistu, and Woldeyes, 2021; Lees et al., 2025), we hone in on how emerging technologies can trigger political backlash as the relationship between voters and the state shifts. This adds to wider literature on the political economy of digitalization across the Global South, from its impact on election campaigns and administration (Cheeseman et al., 2020; Gonzalez, 2021; Garbiras-Díaz and Montenegro, 2022; Fisher, Gadjanova, and Hitchen, 2023; Donati, 2023; Yeandle, 2025b), through to how voters coordinate protest activities and share political information (Christensen and Garfias, 2018; Manacorda and Tesei, 2020; Yeandle, Forthcoming). But despite the ubiquity of mobile money and its profound impact on the technical reach of the state, minimal attention has been paid to its role in taxation and the social contract. We provide some of the first evidence that mobile money taxes can undermine tax morale in emerging economies.

These findings have important implications for policy. Perhaps most pertinent is that by better understanding the dynamics that underpin tax morale, we point toward several ways in which mobile money taxes might be better designed to ensure wider support. For instance, if politicians can make clear why these taxes are necessary, what measures are in place to protect the most vulnerable, and what tangible return citizens will receive through public goods, there is some evidence that the public opinion may be more receptive (Yeandle, 2025a). Governments may also seek to forge cross-party consensus, or include non-partisan bodies in an administrative or monitoring capacity, to reduce concerns about partisan bias and mitigate the risk that policies fail to endure future handovers of power (Ardanaz, Hallerberg, and Scartascini, 2020).

All of this being said, mobile money taxes remain controversial and debate on whether they can raise revenues sustainably is not yet settled. As governments across Africa navigate growing fiscal pressures, from global oil shocks to the withdrawal of foreign donors, these debates will only become more pertinent over time (Jack and Suri, 2014; Aron, 2018; Riley, 2018; Jablonski et al., 2026). Our findings stress the need to pre-empt the political costs of mobilizing digital revenues, to ensure they do not overcast the economic benefits. Otherwise, mobile money taxes risk killing the "golden goose" that made them (Whitehouse, 2021) and being detrimental to wider sustainable development goals.

CRediT authorship contribution statement

Alex Yeandle: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **David Doyle:** Writing – review & editing, Writing – original draft, Visualization,

Public Disclosure Authorized

The World Bank Economic Review, 37(1), 2023, 49–73
https://doi.org10.1093/wber/lhac022
Article

OXFORD

Method Matters: The Underreporting of Intimate Partner Violence

Claire Cullen

Abstract

This paper analyzes the magnitude and predictors of misreporting on intimate partner violence. Women in Nigeria were randomly assigned to answer questions using either an indirect method (list experiment) that gives respondents anonymity, or the standard, direct face-to-face method. Intimate partner violence rates were up to 35 percent greater when measured using the list method than the direct method. Misreporting was associated with indicators often targeted in empowerment and development programs, such as education and vulnerability. These results suggest that standard survey methods may generate significant underestimates of the prevalence of intimate partner violence, and biased correlations and treatment effect estimates.

JEL classification: C83, J12, J16, K42, O15

Keywords: gender, intimate partner violence, measurement, Nigeria

Public Disclosure Authorized

1. Introduction

It is challenging to accurately measure sensitive attitudes, behaviors, and experiences such as political preferences, prejudice, risky sex, and intimate partner violence (IPV). Often, researchers and policy makers must rely on self-reported survey data, yet such data may not be accurate. For sensitive topics in particular, there are many reasons why survey respondents may be reluctant to tell the truth, including a fear of

Claire Cullen is postdoctoral research fellow from the Blavatnik School of Government, University of Oxford; her email address is claire.cullen@bsg.ox.ac.uk. The impact evaluations through which these data were collected were conducted by the World Bank Africa Region Gender Innovation Lab and were funded by the World Bank's Nordic Trust Fund, the Swiss Development Cooperation, and the Government of Rwanda, and in Nigeria by USAID and the Umbrella Facility for Gender Equality, a World Bank Group multi-donor trust fund. The views presented in this paper are those of the author and do not represent those of the World Bank or its member countries. I gratefully acknowledge support from the Blavatnik School of Government, Fund for Women Graduates, and Graduate Women International. I would like to thank anonymous reviewers for their helpful feedback, as well as Julien Labonne, Stefan Dercon, Arthur Alik-Lagrange, Noam Angrist, Dara Kay Cohen, Cheryl Doss, Patricia Fernandes, Markus Goldstein, Harald Hinkel, Eliana La Ferrara, Summer Lindsey, Anandi Mani, Sreelakshmi Papineni, Rachael Pierotti, Simon Quinn, Diana Contreras Suarez, Alex Twahirwa, Julia Vaillant, Marc Witte, and seminar participants at the World Bank Africa Region Gender Innovation Lab, Oxford CSAE, IFPRI, UC Berkeley, Yale EGEN, Australian Gender Economics Workshop, Kigali Impact Hub, and SVRI for helpful comments and suggestions. I would also like to thank the respondents for generously giving their time, and the staff at Laterite Ltd. in Rwanda and TNS Global in Nigeria, and Victoria Isika, Jean Paul Ngabonziza, Maritetou Sanogo, and Oluwatoyin Zakariya for their superb research assistance and data collection work. All remaining errors are my own. This study was approved by the Oxford University Central University Research Ethics Committee, HML IRB, and the Rwandan National Ethics Committee. This paper was previously circulated under the title "Method Matters: Underreporting of Intimate Partner Violence in Nigeria and Rwanda." A supplementary online appendix is available with this article at *The World Bank Economic Review* website.

© The Author(s) 2022. Published by Oxford University Press on behalf of the International Bank for Reconstruction and Development / THE WORLD BANK. This is an Open Access article distributed under the terms of the Creative Commons Attribution License (https://creativecommons.org/licenses/by/4.0/), which permits unrestricted reuse, distribution, and reproduction in any medium, provided the original work is properly cited.

Downloaded from https://academic.oup.com/wber/article/37/1/49/6883824 by International Monetary Fund / World Bank - IMF user on 11 September 2023

being judged, endangered, or legally penalized (Blair, Coppock, and Moor 2020). If misreporting bias is significant and systematic, researchers and policy makers risk over- or underestimating the prevalence and welfare costs of sensitive issues and drawing inaccurate conclusions about their causes and potential solutions.

This paper compares women's self-reported experience of IPV under different survey methods. In a survey experiment in Nigeria, women were randomly assigned to answer questions about their experience of emotional, physical, and sexual IPV using one of two methods: an indirect method (list experiment) or the standard survey method (face-to-face questions asked by an enumerator). Compared to the direct face-to-face method, the list experiment provides additional confidentiality as respondents do not directly disclose their experience of violence. Instead, they give a response that allows researchers to calculate average prevalence across the whole sample, essentially making respondents' answers anonymous (Blair and Imai 2012).

Results show that estimated IPV prevalence varies depending on the survey method used to measure it. The most widely used method, face-to-face interview, resulted in the lowest IPV rates, while the anonymous list method had higher rates.[1] Women's reported experience of emotional and physical IPV was 31 and 35 percent greater when measured using the list method compared to the direct method. However, we find no statistically significant difference between the list and face-to-face methods for sexual IPV. Anecdotal evidence and the similarly high rates of sexual IPV reported for both methods (26 and 28 percent) suggest this lack of difference for sexual violence could potentially be explained by a context and interview setting where reporting forced sexual intercourse within marriage was less stigmatized.

This paper finds some similar patterns of misreporting in a related experiment in Rwanda, where we randomly assigned women and men to answer violence questions using one of three methods: the indirect list experiment and two direct survey methods (face-to-face questions asked by an enumerator, or audio computer-assisted self-interview (ACASI) on an electronic tablet). The ACASI method affords more privacy than enumerator-administered methods as respondents answer questions themselves by listening to pre-recorded questions in a headset and then selecting their answer on a tablet. In Rwanda, as in Nigeria, we find women report more physical IPV when interviewed under the list experiment than the direct methods.[2] In this paper, we primarily focus on the Nigeria experiment as it uses the more common list experiment design and IPV questions based on the DHS. The Rwanda experiment is discussed in the supplementary online appendix.

In addition to significant differences in reporting rates, this paper also finds that misreporting is systematic and correlated with characteristics that have been theorized as causes of IPV and targeted in development and female empowerment programs. The characteristics that correlate differently with IPV depending on the measurement method used include indicators of women's education, vulnerability, and marital relationship quality. For example, literate Nigerian women are 8 percentage points less likely than illiterate women to experience emotional IPV when violence is measured using the face-to-face method, but 12 percentage points more likely if IPV is measured using the list method. This suggests that more-

1 In more than 60 developing countries, Demographic and Health Surveys (DHS) have included a standardized domestic violence module where enumerators directly ask women about their experience of a set of specific emotional, physical, and sexual abuse episodes by an intimate partner. Using the face-to-face method, the WHO estimates that approximately one in three women worldwide has experienced IPV in her lifetime (García-Moreno et al. 2013). IPV prevalence varies widely, though rates are generally highest in Africa (Devries et al. 2013; Global Burden of Disease 2017). Researchers collecting IPV survey data take steps to minimize the risk of underreporting, for example by instituting ethics protocols about privacy during interviews, asking multiple questions to give respondents several opportunities to disclose violence, and using specially trained female enumerators to conduct women's interviews (WHO 2016, 2001; Heise and Hossain 2017).

2 In Rwanda we measured, using the three methods, women's experience of physical IPV and non-partner sexual violence and men's reported use of controlling behavior and emotional IPV.

Downloaded from https://academic.oup.com/wber/article/37/1/49/6883824 by International Monetary Fund / World Bank - IMF user on 11 September 2023

educated women underreport IPV when asked directly but that the list method's anonymity may make them comfortable enough to report it. This accords with results from other contexts (Joseph et al. 2017; Agüero and Frisancho 2022).

Results suggest that the standard, direct methods currently used in most surveys risk generating significant underestimates of IPV prevalence. Further, social scientists risk estimating biased correlations and, if treatment is correlated with misreporting, estimating biased treatment effects and drawing inaccurate conclusions about IPV's causes and the effectiveness of interventions to prevent it.[3]

This paper contributes to the growing social science literature studying alternative survey methods that may reduce misreporting (Karlan and Zinman 2012; Blattman et al. 2016; Blair, Coppock, and Moor 2020; Chuang et al. 2021), including on IPV measurement. Recent studies on IPV measurement have compared respondents' experience of violence across the face-to-face and list experiment methods. They found mixed results, with one study in Peru finding the same rates of IPV between face-to-face and list methods (Agüero and Frisancho 2022) and others in Vietnam, India, and Burkina Faso finding a difference (Joseph et al. 2017; Bulte and Lensink 2019; Lépine, Treibich, and d'Exelle 2020).[4] However, the latter three of these studies asked respondents about their IPV experience using two methods within the same survey about the same or similar, but slightly different, violence acts. This potentially limits comparability across methods as respondents may answer differently the second time they are asked the same question. To avoid this issue, we randomly assign respondents to answer questions using only one method rather than asking respondents two overlapping questions with different methods. This paper also contributes to the limited evidence on who misreports on sensitive questions, the implications of misreporting for bias, and the trade-offs between different survey methods that seek to address misreporting.

We contribute to recent debates about the list experiment's accuracy and efficiency in generating prevalence estimates, and its relevance given evidence that misreporting under direct methods can be small. For example, Chuang et al. (2021) identify challenges with the consistency of the prevalence of sexual and health behaviors estimated using list experiments, finding that prevalence sometimes differed depending on the other questions included on the list. Other questions about the list experiment concern its trade-off between potential bias reduction from minimizing misreporting, and the efficiency of its estimates given it adds noise to the data (Lépine, Treibich, and d'Exelle 2020). Finally, a recent review of reporting on sensitive topics in political science found that the difference in prevalence estimates between list and direct methods is often only around 5 percentage points (Blair, Coppock, and Moor 2020). If also true for IPV, there may be a limited upside to using the method. However, we show that face-to-face underreporting on IPV is likely to be substantial in some contexts. This highlights the continued need to carefully design and test list experiments, while also identifying new methods or interviewing techniques that minimize misreporting, are consistent, precise, and feasible to implement in resource-constrained settings. While the ACASI method holds some promise, it produced broadly similar IPV estimates to the face-to-face method in our experiment in Rwanda.

This paper contributes to the nascent literature comparing IPV prevalence using multiple survey methods that offer respondents different levels of privacy and anonymity. It is also one of few papers to study IPV misreporting in Africa, and to demonstrate that misreporting is substantial in some contexts, varies by

3 A further motivation for assessing IPV measurement methods is that the status quo of face-to-face questions places a high emotional burden on both respondents and enumerators. WHO guidelines thus recommend particular protocols for IPV research, for example ensuring respondents have access to referral services (WHO 2016). If an alternative method was identified that was less biased, logistically easier, and more ethical than the status quo, then it would likely be adopted by more researchers and contribute to new research and evidence on IPV (Peterman et al. 2017).

4 A related study by Agüero et al. (2020) in rural Peru compares women's direct self-reported experience of IPV with a listing of IPV survivors by community leaders. They found that leaders underreport IPV compared to women's self-reports.

Downloaded from https://academic.oup.com/wber/article/37/1/49/6883824 by International Monetary Fund / World Bank - IMF user on 11 September 2023

the type of violence, and is correlated with characteristics commonly targeted in women's empowerment interventions.

2. Measuring Sensitive Topics

There is robust evidence that people underreport the truth about sensitive topics when they are directly asked in a survey. Such research compares self-reported survey answers to observed or independently verifiable outcomes such as tax records, receipt of welfare benefits, drug prescriptions for mental illness, and voting in an abortion law referendum (Gottschalk and Huynh 2010; Rosenfeld, Imai, and Shapiro 2016; Bharadwaj, Pai, and Suziedelyte 2017; Murray-Close and Heggeness 2018; Meyer and Mittag 2019).

However, reliable, unbiased observational data are rare for many sensitive topics, including IPV.[5] This is particularly true in developing countries, where there is limited administrative data from the medical or legal services that IPV survivors may use. Non-survey data sources that address the risk of reporting bias, such as emergency and domestic violence hotline calls, hospital and police administrative records, and Google search data (Leslie and Wilson 2020; Ravindran and Shah 2020; Bullinger, Carr, and Packham 2021), were widely used to study the effect of COVID-19 lockdowns on IPV. However, they typically capture the more extreme cases of IPV and might still be biased in other ways, for example by the types of people selecting into services. For example, in Nigeria, only 1 percent of women who experienced physical or sexual IPV reported it to the police (DHS 2019). While administrative and other non-survey data sources can be useful for studying extreme IPV cases and to validate survey data in some contexts, there is still a need for accurate, representative survey data from lower-resource settings and for less extreme cases.

As a result of these non-survey data limitations and concerns about underreporting on face-to-face surveys, researchers have considered alternative survey methods to minimize misreporting bias. A popular alternative method, the list experiment broadly aims to address misreporting driven by shame, embarrassment, a preference for anonymity, or fear of repercussions if one's answer becomes known.

2.1. The List Experiment

Researchers have used the list experiment to improve data accuracy when measuring sensitive behaviors and attitudes.[6] The standard list experiment works as follows: respondents are randomly assigned to one of two groups to answer the list question. In the "non-sensitive" or "control" group, enumerators read respondents a set of three non-sensitive statements and ask the respondent to tell them how many but not which statements are true for the respondent, with possible answers between zero and three. Meanwhile, the "sensitive" or "treatment" group is read the same list of statements, with the addition of a fourth sensitive statement. They are given the same instructions, to say how many, between zero and four statements, are true. Given the sensitive and non-sensitive list groups are randomly assigned, the two groups are, in expectation, equivalent and the only difference is the inclusion of the sensitive statement. The prevalence of the sensitive item across the whole sample is thus the difference in the mean number of items reported by the two groups.

5 Tourangeau and Yan (2007) define sensitive questions as those a respondent deems intrusive, where they fear the threat of disclosure and the possible consequences of giving a truthful response, or where there are social norms that imply there are socially desirable and undesirable responses.

6 For example, Miller (1984) developed the list method to study illicit drug use. Others have used it to study sensitive topics including prejudice in the United States (Kuklinski, Cobb, and Gilens 1997), political preferences in Lebanon (Corstange 2009), support for militant groups in Afghanistan (Blair, Imai, and Lyall 2014), risky sexual behavior (Jamison et al. 2013; Chuang et al. 2021), support for female genital cutting (De Cao and Lutz 2018), vote buying (Gonzalez-Ocantos et al. 2012), and use of microfinance loans (Karlan and Zinman 2012).

Downloaded from https://academic.oup.com/wber/article/37/1/49/6883824 by International Monetary Fund / World Bank - IMF user on 11 September 2023

Unlike with direct methods, anonymity is ensured because the lists are designed in such a way that very few people in the list sensitive group have all four statements as either true or not true.[7] This way, a respondent is not revealing their true answer to the sensitive item to the enumerator or anyone with data access. This could mitigate underreporting by respondents concerned about shame or consequences if someone were to find out their answers, or those who doubt the confidentiality of their data.

There are potential drawbacks to the list experiment. One challenge is that, by design, it cannot be used to identify individuals who respond affirmatively to the sensitive question of interest, potentially limiting its use where individuals need to be referred to support services. Also, with list experiment answers ranging from zero to four, it typically has greater variance than direct questioning methods (but also likely lower misreporting bias because of respondent anonymity). Thus list experiments may reduce bias at the cost of efficiency (though there are approaches such as the double list experiment that can reduce the method's variance) (Blair and Imai 2012).[8] Given these power concerns, the list method may not be ideal for use in impact evaluations as it may leave outcome analysis, and heterogeneity analysis, underpowered. Further, the method does not easily lend itself to further question complexity, for example follow-up questions to measure frequency and injuries from violence, or aggregation of several questions to create indicator variables of "any experience of violence." Finally, Chuang et al. (2021) find evidence that prevalence estimates under the list experiment can be internally inconsistent, with rates that are sensitive to the other questions used in the list.[9]

However, given the list method's potential to reduce misreporting, researchers have begun using it to measure IPV and other forms of gender-based violence. In several studies, IPV rates are compared between the face-to-face and list methods. The study most related to this paper, conducted in urban Lima, Peru, found no difference in overall IPV prevalence by survey method (Agüero and Frisancho 2022); studies in other contexts did find a difference. In Sri Lanka, Traunmüller, Kijewski, and Freitag (2019) found the self-reported experience of wartime sexual violence (not IPV) to be higher under the list method than face-to-face, and in India and Burkina Faso, Joseph et al. (2017) and Lépine, Treibich, and d'Exelle (2020), respectively, found higher rates of IPV under the list method than face-to-face.[10]

In these studies, respondents were either (a) randomly assigned to answer questions about the same experience of IPV using the direct face-to-face method or the list method (Agüero and Frisancho 2022), or (b) all respondents were asked about their IPV experience using both methods within the same survey—either with similar, but slightly different, question wording (Bulte and Lensink 2019) or the same question wording (Joseph et al. 2017; Lépine, Treibich, and d'Exelle 2020).[11] One issue with approach (b) is that

7 This can be done through careful piloting and selecting a set of statements that are all plausible but where the co-occurrence of all being true or untrue for any one person is unlikely. See Glynn (2013) and Corstange (2009) for a discussion and suggestions.

8 Droitcour et al. (1991) proposed the double list experiment, where all respondents participate in two list experiments with different non-sensitive items but the same sensitive item. The combined estimate across the two experiments has approximately half the variance of the standard list experiment. Other IPV list experiment trade-offs are discussed in the supplementary online appendix.

9 The authors implemented several double list experiments to measure sexual behavior in Cote d'Ivoire. They found that prevalence sometimes differed between the two lists, potentially because different lists violated key list experiment assumptions.

10 Peterman et al. (2017) measured physical violence prevalence using the list method in Zambia. They found rates of IPV similar to those reported in the Zambian DHS (conducted face-to-face), though the samples are not directly comparable.

11 For example, in Bulte and Lensink's (2019) Vietnam study, the sensitive list method group responded to a list that included the sensitive statement "I am regularly hit by my spouse," and then all respondents later in the survey were also asked, "How often did your husband push, slap, beat or hit you during the last 6 months," with response options never, rarely, sometimes, often, very often, or refuse. With this design, the questions may not be directly comparable given potential differences in interpretation and biased responses to the second question. Similarly, Joseph et al. (2017) first asked all respondents using the list method, with the sensitive statement being "At least one woman member of my

Downloaded from https://academic.oup.com/wber/article/37/1/49/6883824 by International Monetary Fund / World Bank - IMF user on 11 September 2023

being asked the same question in two ways may bias responses to the second question. In our Nigeria experiment, we take approach (a), asking individuals each question only once to minimize this risk.

2.2. Misreporting and Truth

Tourangeau and Yan (2007) argue that for sensitive questions where responding in the affirmative is taboo, the survey method that finds the highest prevalence is closest to truth. In this paper, particularly given the stigma surrounding the issue of IPV, we assume that in most cases, the method with the highest prevalence is closest to truth and that most misreporting is underreporting.

Without objective, non-self-reported data, we cannot, ultimately, confirm which method is closest to truth. However, the alternative survey methods tested in this paper are designed to address motivations for underreporting when interviewed with the face-to-face method—shame and fear. These alternative methods are unlikely to introduce new reasons for people to report IPV that they have not experienced. This is particularly true for the list experiment, where answers are anonymous by design, hence individuals have no incentive to overreport given they cannot be followed up with to provide aid or program services.[12] Therefore, it seems likely that the method with the highest reporting indicates the method that best addresses respondents' reasons for misreporting and is closest to truth.

This assumption is supported by empirical evidence from the only study we identified that compares an observed sensitive behavior with self-reported behavior measured using direct and indirect methods including the list experiment. Rosenfeld, Imai, and Shapiro (2016) found that when they were asked directly, people underreported their past voting behavior on a sensitive abortion referendum in Mississippi. However, rates were higher and closer to truth when they were asked using the list experiment. No method yielded rates higher than the true "socially undesirable" outcome of voting against tightened abortion restrictions.

Overall, evidence on the measurement of sensitive topics suggests that misreporting appears to be highly issue and context dependent (Tourangeau and Yan 2007; Langhaug, Sherr, and Cowan 2010; Blair, Coppock, and Moor 2020), presumably affected by factors such as the interview setting, cultural norms, security and safety, enumerator skill, and confidence in data privacy. Therefore, this paper primarily relates to misreporting about IPV, and in contexts similar to those studied here.

3. Experimental Design

3.1. Context, Data, & Design

Countries in Sub-Saharan Africa have high rates of IPV compared to the global average. According to data gathered using the direct face-to-face method, 14 percent of Nigerian women report having experienced physical or sexual IPV in the previous 12 months (DHS 2019). This compares to 3 percent of women in the United States, 21 percent in Rwanda, 23 percent in Kenya, 18 percent in India and Bolivia, and 10 percent in South Africa (Global Burden of Disease 2017; DHS 2021).

Our study took place in one of Nigeria's poorest and most conservative regions, Kebbi state in Northwest Nigeria. Kebbi has the country's highest acceptance of wife-beating, with 67 percent of women believing a husband is justified in using physical violence if a wife burns the food. Yet it also has the country's lowest reported rates of IPV (DHS 2019). Only 13, 7, and 1 percent of women report experiencing emotional, physical, and sexual IPV in the previous 12 months (compared to the national average

household has faced physical aggression from her husband anytime during her life," followed by all respondents directly being asked, "Has at least one woman member of your household faced physical aggression from her husband anytime during her life?'

12 However, it is also plausible that by presenting the item of interest in a list of other items, the list method affects the way people interpret the severity of violence and/or differentially affects recall, introducing a different type of reporting bias.

Downloaded from https://academic.oup.com/wber/article/37/1/49/6883824 by International Monetary Fund / World Bank - IMF user on 11 September 2023

of 26, 12, and 5, respectively). In this region, IPV is not criminalized. The penal code governing Northern Nigerian states does not outlaw marital rape or physical violence against a wife "so long as the woman is not seriously harmed" (Benebo, Schumann, and Vaezghasemi 2018).[13]

The survey experiments described in this paper were conducted in accordance with WHO guidelines on the safe and ethical collection of data on violence against women (WHO 2001, 2016).

In Nigeria, the measurement experiment was conducted during the endline survey of a cluster randomized control trial studying the effects of the Feed the Future Nigeria Livelihoods Project, a multicomponent cash transfer and livelihoods program.[14] The sample consisted of ultra-poor households, with approximately 15 percent of women self-reporting as literate, 30 percent in polygamous marriages, and 84 percent identifying as Muslim.

The eligibility criteria for the measurement experiment were that the respondent was female, over 17 years of age, and married and living in the same household as a male. In total, 2,817 women met these criteria and were randomly assigned to answer three experimental questions about their experience of IPV using either the face-to-face or list method. Assignment to the method groups was stratified by geographic and poverty-related strata, as well as treatment status in the randomized control trial, to ensure balance.

The three IPV questions used for the measurement experiment were either directly from or combined several violence questions from the DHS.[15] The experimental questions were "In the past 12 months, has your husband

(1) said or done something to humiliate you in front of others?
(2) slapped you, pushed you, shaken you, or thrown something at you?
(3) physically forced you to have sexual intercourse with him when you did not want to?"

Respondents were interviewed one-on-one by a female enumerator in a private setting, usually in the household compound. Enumerators were instructed to pause if anyone came within earshot of the interview, not proceeding until privacy was ensured. The survey structure is outlined in table A.1.

3.2. The Measurement Experiment: Calculating Prevalence

We assess misreporting bias by comparing rates of IPV across the face-to-face and list experiment methods (with a third method, ACASI, also tested in Rwanda and discussed in the supplementary online appendix). Given that assignment to the survey method was random, we assume that the only difference in reported violence between the groups is due to the method used.

Under the list method, prevalence of the sensitive statement is calculated by taking the difference in the mean number of statements reported as true by the two groups (Blair and Imai 2012). To administer the list experiment to the non-sensitive group, respondents are read the same list, bar the fourth sensitive statement, and asked to report how many of the three statements apply to them.[16]

13 Cases of domestic violence are hardly ever brought to trial as law and justice agents typically treat IPV as a family matter, and police often fail to respond if they consider cases to be within cultural norms (Benebo, Schumann, and Vaezghasemi 2018). For the very few violence survivors choosing to report or leave their husband (DHS 2019), limited official resources are available.

14 Data were collected from May–July 2018, with the program ending 12 months prior (Bastian, Goldstein, and Papineni 2017). The study and experiment were pre-registered on the American Economic Association trial registry.

15 Married female respondents were first asked about their experience of IPV using the list method (asked either the sensitive or non-sensitive list, depending on her randomly assigned group), followed by the three direct face-to-face IPV questions if she was in the list non-sensitive group.

16 To ensure innumerate respondents could answer list questions, and to help all respondents keep track of their answers, they were advised to put their fist behind their back or under their hijab out of sight of the enumerator and to straighten a finger each time a true statement was read out. Once all statements had been read out, respondents showed the enumerator their hand to count the number of straightened fingers. Some respondents also put their hands behind their

Downloaded from https://academic.oup.com/wber/article/37/1/49/6883824 by International Monetary Fund / World Bank - IMF user on 11 September 2023

Prevalence under the list method was estimated using an OLS regression as

$$Y_i = \alpha + \rho T_i + \epsilon_i,$$

where ρ is prevalence, or the difference in the mean number of items reported in the groups, Y_i is the number of statements on the list that are true for individual i, α is the mean number of items for the non-sensitive group, and T_i is assignment to the list treatment group.

In Nigeria, 1,405 women were randomly assigned to answer the sensitive questions using the list method group, and 1,412 using the face-to-face method.[17] Power calculations indicated that the sample in Nigeria was large enough to detect a difference of approximately 6 percentage points between the groups, and in Rwanda of approximately 5 percentage points, with a significance level of 0.05, power of 0.8, and base prevalence of 20 percent.[18]

Overall, the groups were well balanced on key characteristics, as shown in table A.3. While several of the raw differences are statistically significant at conventional levels, the differences are small, and they are within the bounds of what might be expected given the number of variables tested for balance. No normalized differences are greater than 0.25 standard deviations, a suggested rule of thumb indicating good balance (Imbens and Rubin 2015). In additional specifications, we estimate prevalence including controls selected using post-double selection LASSO, and variables with baseline imbalance (Belloni, Chernozhukov, and Hansen 2014).[19] These do not significantly change results.

4. Results

4.1. Method Matters: IPV Reporting Differs by Survey Method

This paper assesses whether there is a difference in reported rates of IPV across survey methods. A difference in prevalence implies misreporting under at least one of the methods.

Table 1 shows IPV prevalence by survey method in Nigeria. Column 3 shows the difference in prevalence between methods. The p-values for these prevalence difference tests were adjusted for multiple hypothesis testing. Sharpened q-values were computed using the approach proposed by Benjamini and Hochberg (1995) and Anderson (2008).[20]

Results show that the survey method used can substantially affect IPV reporting. Overall, the most widely used survey method, face-to-face interviewing, resulted in the lowest reported IPV rates, with the list method having higher rates. In Nigeria, emotional and physical IPV rates were 31 and 35 percent greater, respectively, when measured using the list method (emotional violence: 39.2 percent; physical

backs and transferred a stone from one hand to the other each time a true statement was read out.
The non-sensitive questions that were asked in the list experiments are shown in table A.2. They were selected from a larger pool of piloted questions following design advice discussed in Glynn (2013), and informed by questions used by Peterman et al. (2017) and Agüero and Frisancho (2022).

17 Differences in prevalence estimates across the methods were calculated using the "Suest" command in Stata, used to compare "seemingly unrelated estimations." Suest combines the stored results from the different estimations into one parameter vector and simultaneous covariance matrix, allowing the testing of cross-model hypotheses.

18 See Blair, Coppock, and Moor (2020) for a comprehensive discussion on list experiment power considerations.

19 Post-double selection LASSO controls for covariates selected by LASSO to predict the outcome. It allows for the parsimonious and consistent selection of controls from a large set of potential variables. It estimates two LASSO regressions, first with the outcome as dependent variable and control variables as regressors, second estimating a LASSO with the treatment as the dependent variable and again the controls as regressors. This second LASSO regression essentially selects unbalanced variables as controls. The final choice of control variables in the OLS regression of the outcome on the treatment indicator includes the controls selected from both LASSO regressions. We use the Stata command "pdslasso" to implement this procedure.

20 The q-values control for the false discovery rate for the three tests for difference in means that we calculate in Nigeria. All regression estimates were calculated with heteroskedastic-consistent robust standard errors.

Downloaded from https://academic.oup.com/wber/article/37/1/49/6883824 by International Monetary Fund / World Bank - IMF user on 11 September 2023

Table 1. Prevalence of Intimate Partner Violence by Survey Method in Nigeria

	Difference between pairs of methods		
	(1) Face-to-face mean	(2) List method prevalence (difference in means)	(3) List−F2F (3)−(1)
In the past 12 months, has your husband			
Q1: said or done something to humiliate you in front of others	0.299***	0.392***	0.093**
	(0.012)	(0.031)	[0.013]
N	1,394	2,817	
Q2: slapped you, pushed you, shaken you, or thrown something at you	0.192***	0.259***	0.067*
	(0.011)	(0.035)	[0.055]
N	1,394	2,817	
Q3: physically forced you to have sex when you did not want to	0.281***	0.261***	−0.020
	(0.012)	(0.032)	[0.551]
N	1,381	2,817	

Source: Data from author's experiment.

Note: The left column shows the violence variables. Columns 1 and 2 show prevalence by method. Robust standard errors are shown in parentheses. Column 3 shows differences in prevalence between the methods. For the test of difference in means, Benjamini and Hochberg (1995) adjusted *q*-values are displayed in square brackets, from a chi-squared test of the differences in means across groups. Regressions are adjusted for stratification. f2f, face to face. ***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

violence: 25.9 percent) compared to face-to-face (emotional violence: 29.9 percent; physical violence: 19.2 percent). The differences between the methods were statistically significant at the 5 and 10 percent levels respectively. However, there was no statistically significant difference between direct and indirect rates of sexual IPV, with reported prevalence of 28.1 and 26.1 percent respectively.

The second experiment, conducted in Rwanda and discussed in the supplementary online appendix, shows a similar pattern of results. There, the third method tested, ACASI, produced similar or slightly higher rates of sexual and intimate partner violence experience (for women) or perpetration (for men) than the face-to-face method.

Our rates of directly reported IPV are much greater than equivalent rates from Nigeria's DHS, shown in table 2. For example, in the DHS, only 15.2 percent of women say their partner said or did something to humiliate them in front of others; 9.2 percent of women say they have been slapped; and 4.8 percent say they have been pushed, shaken, or had something thrown at them in the previous 12 months by their partner (DHS 2019). Four percent say that in the past 12 months, they have been physically forced to have sexual intercourse with their partner when they did not want to. While the DHS is conducted in a different, representative sample and thus is not directly comparable to our targeted, highly economically vulnerable sample, the contrast in IPV rates with both our direct and list experiment estimates is striking. Our much greater face-to-face rates than those in the DHS may be due to our different sample, our very private interview setting, or our two-hour long survey covering many other non-sensitive issues that gives more plausible deniability to respondents and allows enumerators to build a strong rapport before asking IPV questions.

Our results are comparable to those from several related studies. Lépine, Treibich, and d'Exelle (2020) find significant face-to-face underreporting of physical IPV in rural Burkina Faso (16–20 percentage point difference between face-to-face and list method prevalence), as do Bulte and Lensink (2019) in Vietnam (10 percentage point difference), and Joseph et al. (2017) in India (9 percentage points). However, our results diverge from those of the one other study that also adopts our one-method-per-person approach. In

Downloaded from https://academic.oup.com/wber/article/37/1/49/6883824 by International Monetary Fund / World Bank - IMF user on 11 September 2023

Table 2. Comparison of Women's Reported Experience of IPV in the Previous 12 Months by Method and Source (in Percent)

	Survey experiment			
	Face-to-face	List experiment	ACASI	Nationally representative DHS data
		Nigeria		
Emotional IPV				
"Partner said or did something to humiliate you in front of others"	29.9	39.2	—	15.2
Physical IPV				
"Partner has slapped you, pushed you, shaken you, or thrown something at you"	19.2	25.9	—	"Partner has slapped you": 9.2;"Partner has pushed you, shaken you, or thrown something at you": 4.8
Sexual IPV				
"Partner physically forced to have sexual intercourse with your partner when you did not want to"	28.1	26.1	—	4
		Rwanda		
Physical IPV				
"Partner pushed you or thrown you against something"	9.3	20.6	8	"Partner has pushed you, shaken you, or thrown something at you": 9.8

Source: DHS (2019), DHS (2021) and author's own data.
Note: This table shows IPV prevalence from this survey experiment by method, and compares this to IPV rates reported face-to-face in nationally representative Demographic and Health Surveys in Nigeria and Rwanda. Reported in percent. DHS, Demographic and Heath Survey; ACASI, Audio computer assisted self-interview

Lima, Peru, Agüero and Frisancho (2022) found no overall difference between face-to-face and list method physical and sexual IPV prevalence. These divergent results indicate that reporting bias is, unsurprisingly, strongly affected by contextual factors. Differences in misreporting may be due to the social norms and degree of stigma women faced in each context (e.g. urban Lima versus rural Nigeria), or differences in the interview context (e.g. privacy of the interview setting or respondents' faith in the privacy of their data). Significant reporting differences have also been found in other surveys, for example underreporting of condom use or sexual assault during war (Jamison et al. 2013; Traunmüller, Kijewski, and Freitag 2019), suggesting that sexual and health topics in particular may be prone to reporting bias; the direct method–list method differences appear to be much smaller in many political science studies (Blair, Coppock, and Moor 2020).

Overall, our results suggest that women's experience of emotional and physical IPV may be substantially underreported when measured using direct methods. If the difference in prevalence between the list and direct methods is a proxy for fear, shame, or social desirability bias, then these results also suggest that questions about emotional and physical IPV were more sensitive than sexual IPV in our setting. This surprising result could potentially be explained by the context. In Nigeria, a quarter of the women interviewed using the list method and a quarter interviewed using the face-to-face method reported experiencing sexual IPV in the past year. Qualitative interviews and focus groups suggested that there may be less stigma associated with reporting sexual IPV than physical IPV. Participants in our predominantly Muslim study sites indicated that religious leaders cautioned male followers not to use physical violence against their wives, but that sexual IPV was viewed differently and was not considered violence. Further, DHS statistics show that women in this region have an extremely high acceptance of violence related to

Downloaded from https://academic.oup.com/wber/article/37/1/49/6883824 by International Monetary Fund / World Bank - IMF user on 11 September 2023

sex, in particular. In Kebbi state's statistics on acceptance of wife beating, the worst in the country, 87 percent of women believe it is acceptable for a man to beat his wife if she refuses to have sex with him, while 67 percent believe the same if she burns the food (DHS 2019). This very widespread belief that husbands are entitled to sex from their wives suggests that sexual violence may not be particularly stigmatized in this context and therefore women may feel as comfortable reporting it face-to-face as with the list method. The more normalized reporting of sexual IPV may also be a coping mechanism in a context where this form of IPV is widespread and tolerated. A potential alternative explanation could be that some might consider sexual violence as resulting from a husband's passion rather than anger, potentially making it less shameful to report than physical or emotional violence, which can be seen as disciplining errant wives (Benebo, Schumann, and Vaezghasemi 2018).

In the study setting, IPV, including marital rape, is not criminalized. It seems likely that the different legal, religious, and social treatment of emotional, physical, and sexual IPV may contribute to their relative reporting sensitivity.

4.2. Characteristics of Misreporting

In this section, we briefly consider whether misreporting is random, and if not, identify the characteristics that predict it. To do this, we assess correlations between respondent characteristics and IPV, as reported under each of the survey methods. When the IPV-characteristic coefficient differs across methods, this implies systematic misreporting under at least one of the methods that is associated with the characteristic. This analysis provides suggestive evidence on the type of person that might misreport, potential reasons for misreporting, and the risks of drawing biased inferences from such data.

4.2.1. Estimation: Characteristics Associated with Misreporting

For those assigned to the face-to-face group, we assess the characteristics of those reporting IPV by regressing women's self-reported experience of IPV against the variable of interest, such as years of education, as

$$Y_i = \alpha + \delta x_i + \epsilon_i.$$

For the list method, we regress the number of list items reported by an individual against an indicator variable for whether they were in the list sensitive group, as well as the variable of interest and the interaction term. This regression model is

$$Y_i = \alpha + \rho T_i + \gamma x_i + \delta(T_i \cdot x_i) + \epsilon_i,$$

where Y_i is the number of true statements for individual i, T_i is assignment to the sensitive list group, x_i is the characteristic, and δ is the parameter of interest.[21] If there is no (or consistent) misreporting across the methods, the relevant coefficients for each method should be equal. If there are differences, this implies that the relationship between IPV and the characteristic is sensitive to the measurement method, and we may be estimating biased descriptive and causal relationships.

4.2.2. Results: Respondent Characteristics Predict Misreporting

We analyze misreporting by respondent characteristics that fall into three overlapping categories. These characteristics include (a) those most likely to motivate respondents to misreport (to understand why people misreport IPV), (b) variables typically targeted in IPV prevention interventions (to evaluate risks of estimating biased treatment effects), and (c) characteristics frequently cited as IPV risk factors. The first two categories of characteristics include perceptions of village gender norms, poverty and vulnerability, marital relationship quality, and personal gender attitudes. Additional risk factors considered include

21 This approach follows that of Holbrook and Krosnick (2010), Coutts and Jann (2011), and Agüero and Frisancho (2022).

Downloaded from https://academic.oup.com/wber/article/37/1/49/6883824 by International Monetary Fund / World Bank - IMF user on 11 September 2023

Table 3. Correlations between IPV and Respondent Characteristics across Survey Methods

	Q1: emotional violence		Q2: physical violence		Q3: sexual violence		Method differences: *p*-values from tests of differences		
	(1)	(2)	(3)	(4)	(5)	(6)	(1)–(2)	(3)–(4)	(5)–(6)
	List	F2F	List	F2F	List	F2F	Q1	Q2	Q3
Variable									
Vulnerability index	0.02	−0.08***	0.09	−0.05***	0.04	−0.05**	0.13	0.04	0.23
	(0.06)	(0.02)	(0.07)	(0.02)	(0.07)	(0.02)			
Unequal-village-gender-norms index	0.05	0.08***	0.09*	0.05***	0.08*	0.05***	0.55	0.44	0.55
	(0.04)	(0.02)	(0.05)	(0.02)	(0.04)	(0.02)			
Relationship-quality index	−0.07	−0.21***	−0.02	−0.21***	−0.05	−0.27***	0.11	0.04	0.01
	(0.08)	(0.03)	(0.09)	(0.02)	(0.08)	(0.03)			
Public-speaking-confidence index	−0.02	−0.08***	0.00	−0.03***	0.00	−0.01	0.12	0.39	0.69
	(0.04)	(0.01)	(0.04)	(0.01)	(0.04)	(0.01)			
Progressive-gender-attitudes index	−0.12*	−0.13***	−0.23***	−0.09***	−0.16**	−0.16***	0.92	0.10	1.00
	(0.07)	(0.03)	(0.08)	(0.03)	(0.07)	(0.03)			
Bargaining power measures									
Wife's age	0.00	0.00**	0.00	0.00	0.00	−0.00	0.57	0.49	0.59
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)			
Wife reads and writes	0.12	−0.08**	0.00	−0.12***	0.11	−0.04	0.03	0.19	0.13
	(0.08)	(0.03)	(0.10)	(0.02)	(0.09)	(0.03)			
Monogamous	0.07	−0.04	0.16**	−0.03	0.12*	−0.05*	0.12	0.01	0.02
	(0.06)	(0.03)	(0.07)	(0.02)	(0.07)	(0.02)			
Polygamous: wife number (1–4)	−0.03	0.02	0.06	0.03	0.13	−0.02	0.56	0.76	0.13
	(0.08)	(0.04)	(0.10)	(0.03)	(0.09)	(0.03)			
Wife brought assets to marriage	−0.04	−0.18***	0.04	−0.13***	−0.02	−0.09***	0.04	0.03	0.33
	(0.06)	(0.02)	(0.07)	(0.02)	(0.07)	(0.02)			

Source: Author's analysis based on own data.

Note: Robust standard errors are shown in parentheses. Regressions are adjusted for sampling stratification. Columns 1–6 show the relevant covariate coefficients from separate regressions. Columns 7–9 show the *p*-values from a chi-squared test of the differences between these coefficients. The vulnerability index includes measures of food poverty and household economic insecurity, and wife's stress. The unequal-village-gender-norms index includes beliefs about unequal gender attitudes held by others in the village. The relationship-quality index includes measures of trust, satisfaction, empathy, and communication in the marriage. The public-speaking-confidence index includes measures of willingness to speak up in public. The progressive-gender-attitudes index includes measures of personal views on the acceptability of IPV and on gender roles. IPV, Intimate partner violence; F2F, face to face. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

women's education and literacy, age, labor-force participation, and other indicators of women's bargaining power, where theory and empirical evidence are mixed on the relationship with IPV (Eswaran and Malhotra 2011; Bobonis, González-Brenes, and Castro 2013; Heise and Kotsadam 2015; Buller et al. 2018; Erten and Keskin 2018; Haushofer et al. 2019).

Table 3 shows the relationships between IPV and respondent characteristics by survey method. Results show several differences in coefficients across the methods, implying that a number of characteristics are associated with systematic misreporting of IPV.[22] These results are also shown graphically in fig. A.1. For ease of interpretation, in table A.4 we also show the differences in mean IPV prevalence for respondents above and below the median of the characteristic, or by their indicator variable characteristic. However, given we are likely underpowered to detect the differences in this table, results should be interpreted with caution.

22 Results are essentially unchanged when we include control variables in the list estimate regressions, including polygamous marriage, region, husband's alcohol consumption, religion, wife earning a wage income, household recently faced a negative income shock, and wife's age at marriage.

Downloaded from https://academic.oup.com/wber/article/37/1/49/6883824 by International Monetary Fund / World Bank - IMF user on 11 September 2023

This subsection presents the results of descriptive analysis and, as such, the usual caution should be applied when interpreting results given that omitted variables or other sources of bias may be driving observed correlations.[23]

Our results suggest that vulnerable women were more likely to report physical IPV under the list method than face-to-face. For emotional and sexual IPV, the same pattern existed but the differences were not statistically significant.

Greater marital relationship quality, measured using questions about satisfaction, empathy, and trust in marriage, was also associated with underreporting. We found that women who reported greater relationship quality were more likely to report IPV when interviewed using the list method than face-to-face; the method differences were statistically significant on two out of the three IPV questions considered. A potential reason for women who report being in a more trusting, supportive relationship to underreport IPV when asked directly is because the direct question forces her to confront an uncomfortable truth about her marriage, or she may feel disloyal by directly admitting her husband has been violent.

These results highlight a potential measurement bias issue for the growing number of policies being assessed on their IPV impact. For example, an intervention that improves relationship quality, as several IPV prevention programs seek to do, may have no impact on actual IPV rates. However, it may lead to a reduction in a wife's willingness to directly report IPV for the reasons discussed above. This would bias treatment effect estimates, suggesting a reduction in IPV that was, in fact, an artefact of the survey method used.

We also examined whether common correlates of IPV remain stable across the method used. For some indicators of women's intrahousehold bargaining power, we find some evidence of systematic misreporting of IPV, though no clear pattern across different indicators.

Concerningly, we observe some opposite-signed relationships, where the sign on the relationship between IPV and the respondent's characteristic is positive for one method and negative for another. In Nigeria, these variables include wife's vulnerability, literacy, and assets brought into the marriage. For example, we find suggestive evidence of a positive relationship between IPV and women's education when IPV is measured directly, but a negative relationship when using the list method. We observe a similar relationship in Rwanda. This is also similar to results in Peru and India, where researchers found women's education to be positively correlated with IPV when measured using the list method but negatively when measured face-to-face (Joseph et al. 2017; Agüero and Frisancho 2022). The opposite-signed IPV–education associations would imply opposing theories about women's bargaining power as a cause of IPV.[24]

Overall, the sensitivity of the relationships between IPV and several bargaining power and other indicators to the survey method used suggests we should be cautious when using direct IPV measures to understand descriptive and causal relationships, as misreporting may bias correlations and treatment effect estimates. The anonymity of the list experiment may minimize this reporting bias, though at the cost of precision.

23 For conceptually linked variables (for example, questions about relationship quality such as trust and empathy in the marriage), we follow Anderson (2008) and generate weighted standardized summary indices as the most efficient weighted average of the set of outcomes to minimize the false discovery rate. The index weights outcomes using the inverse of their variance–covariance matrix. For other indicators of interest where there is less theoretical coherence between the variables, we report coefficients on the raw variable.
Given that this is exploratory analysis only, we do not adjust the p-values of the tests of equivalence for multiple hypothesis testing, though when we do compute q-values to control for the false discovery rate, none of the list–face-to-face differences remain statistically significant at conventional levels. This suggests results should be interpreted with caution.

24 However, we cannot rule out that literate women understood the list experiment better than illiterate women, and thus how much weight to place on these results.

Downloaded from https://academic.oup.com/wber/article/37/1/49/6883824 by International Monetary Fund / World Bank - IMF user on 11 September 2023

4.3. Robustness Checks

In this section we report the results of robustness checks on the list method, followed by checks on the overall analysis.

4.3.1. Checks on List Method Assumptions

There are two assumptions that underpin the validity of statistical analyses of list experiments, and which, if they hold, should result in an unbiased difference-in-means estimator (Blair and Imai 2012). The first assumption is that of "no design effect," meaning that the inclusion of the sensitive item in the list does not affect answers to non-sensitive items. The second assumption is of "no liars," or that the answers given for the sensitive item are true.

Blair and Imai (2012) propose a likelihood ratio test to assess the probability that the first assumption is violated. The test is based on the assumption that the inclusion of a sensitive statement should not change the sum of affirmative answers to the non-sensitive statements, and that cumulative proportions of different respondent types should be non-negative (Bulte and Lensink 2019). If one of the proportions is negative, then the no-design-effect assumption has likely been violated. The rejection of the null hypothesis of no design effect using Blair and Imai's statistical test would suggest there was a design effect. The test returns a Bonferroni-corrected minimum p-value of the test (Blair and Imai 2011).[25]

In Nigeria, the test returned a minimum p-value of 1, suggesting there was no design effect for the three list questions.[26]

There is no explicit test for the second "no liars" assumption. However, the most obvious threat to this assumption is when a respondent's anonymity for their answer to the sensitive statement is compromised. This "ceiling effect" exists when a respondent's true answer is affirmative for all of the statements, thus their affirmative response to the sensitive statement will no longer be hidden. The proportion of respondents answering with the maximum number of items (three and four for the non-sensitive and sensitive list groups, respectively) is within the range of other studies.[27] The presence of a ceiling effect likely indicates downwards-biased list estimates because other respondents would be likely to have understated their true number of affirmative answers to conceal their sensitive answer (Blair and Imai 2012).

4.3.2. Checks on Overall Approach

The two key assumptions underpinning the unbiasedness of the list experiment estimator have parallels for the experiment as a whole.

First, we checked balance on survey questions asked of all respondents after the experiment questions. If respondents were differentially affected by one survey method, perhaps indicating a design effect, we may observe them answering subsequent survey questions differently. We found very few statistically significant differences in answers to subsequent survey questions despite the many tests for significance, and any differences were small.

25 Chuang et al. (2021) found that having relatively sensitive "non-sensitive" statements on the list appears to mask the sensitive item of interest, and results in lower variance and higher prevalence estimates. We used relatively innocuous non-sensitive list items, which, if results from the Chuang et al. measurement experiment on risky sexual behavior hold, make it plausible that our list experiment estimates may be downwards biased.

26 The associated test p-values and proportions for Nigeria are shown in table A.5. In Rwanda, where the list non-sensitive group was directly asked each non-sensitive question, there is statistically significant evidence of a design effect for three out of the four questions, suggesting that group assignment affected answers to the list questions. There was no design effect for the first men's question.

27 The proportion of respondents in the non-sensitive list group reporting three items as true is never greater than 14.7 percent (for the women's physical violence question), and the proportion of respondents in the sensitive list group reporting the full number of items is never greater than 5.4 percent (for the women's physical violence question). Table A.6 simulates a possible upper bound on the physical IPV list estimate if there had been no ceiling effect and under extreme assumptions of underreporting with the sensitive list.

Downloaded from https://academic.oup.com/wber/article/37/1/49/6883824 by International Monetary Fund / World Bank - IMF user on 11 September 2023

Second, we analyzed the main list method prevalence regressions with enumerator fixed effects, controls for pre-experiment imbalance, and controls selected using post-double-selection LASSO (Belloni, Chernozhukov, and Hansen 2014). Estimates remained consistent, with few differences in coefficients or significance, as shown in table A.7.

Third, although a general "no liars" assumption is untestable, we compared refusal rates across methods to indicate the potential relative risk of misreporting. We find very low refusal rates, the highest being 0.64 percent for the face-to-face sexual violence question. Refusal rates are shown in table A.8.

Additional indications that these results are robust are supported by the broadly similar patterns of misreporting by survey method observed over different violence types, countries, and list method designs, and in Rwanda, different genders, and perpetration and victimization questions.

5. Conclusion

This paper provided some of the first evidence that the prevalence of intimate partner violence can vary substantially depending on the survey method used to measure it. By randomizing individuals to answer using only one method, we avoid some pitfalls from earlier studies. Our results suggest that in some contexts, standard survey methods are likely to result in significant underestimates of the prevalence and welfare costs of IPV. The most widely used method, face-to-face interview, resulted in the lowest prevalence estimates, followed by the more private ACASI method (tested in Rwanda). Finally, the list method, which allows respondents to report anonymously, resulted in the highest prevalence. In Nigeria, women's experience of physical IPV was 35 percent greater when measured using the list method than the direct method, and 180 percent greater than physical IPV rates reported in Nigeria's DHS.

This paper also showed that misreporting was systematic and was correlated with indicators that theory suggests are risk factors for IPV and those often targeted in women's empowerment programs, including vulnerability, marital quality, and education.

As other studies have found, direct measures of IPV can produce biased treatment effect estimates, for example, finding programs caused a reduction in IPV if measured face-to-face, but an increase if measured with the list experiment (Bulte and Lensink 2019). Given the growing number of studies measuring program impacts on IPV, our results highlight the potential risk of collecting measurement-error-prone data in generating unbiased correlations and treatment effect estimates.[28]

Our finding of significant and systematic underreporting for some types of IPV in our rural African contexts contrasts with findings from Lima, Peru, where researchers found no overall difference between face-to-face and list method prevalence (Agüero and Frisancho 2022), and with research on sensitive topics in political science, which typically find lower levels of misreporting (Blair, Coppock, and Moor 2020). However, our findings are consistent with IPV misreporting observed in rural Burkina Faso, Vietnam, and India (Joseph et al. 2017; Bulte and Lensink 2019; Lépine, Treibich, and d'Exelle 2020), and in other sensitive health and sexual behavior surveys (Jamison et al. 2013; Traunmüller, Kijewski, and Freitag 2019). Overall, our results indicate that misreporting bias is, unsurprisingly, strongly affected by contextual factors, but can be substantial, and varies by type of violence.

How then should we measure IPV, given our results suggest that IPV is likely to be underreported when measured using the status quo face-to-face method? Policy makers and social scientists measuring sensitive topics could use similar randomized survey experiments to compare IPV reporting rates between the list experiment and direct methods to identify the magnitude of stigma, the types of people most likely to misreport, and the risk of generating biased treatment effects. For example, social scientists researching

28 For example, several randomized control trials of IPV prevention or gender-norms-related interventions have found small or null effects on IPV while having impacts on related outcomes such as gender equitable attitudes or perceptions of norms (Vaillant et al. 2020). It is plausible that such gender norms or couples-focused interventions had an impact on IPV, for good or bad, but that biased reporting producing biased treatment effects may not show this.

Downloaded from https://academic.oup.com/wber/article/37/1/49/6883824 by International Monetary Fund / World Bank - IMF user on 11 September 2023

the impact of interventions on IPV might use such an experiment to sign potential reporting bias in their treatment effect estimates. Also, given DHS's large sample sizes and use in multi-country analysis, perhaps such surveys should measure several IPV questions using the list method as well as the face-to-face method. This would allow data users to assess the likely extent of reporting bias by context, allowing them to make adjustments to inferences about true levels and correlates of IPV.[29]

While the list experiment may reduce reporting bias, the method presents some analytical and administration challenges that may undermine its suitability for measuring IPV in some circumstances, and suggests the need for research into alternative approaches that also reduce misreporting. The larger variance associated with list experiment prevalence estimates makes it difficult to use as an outcome measure in smaller studies. Further, it is more time consuming to administer and requires patient, well-trained enumeration. Finally, there are ongoing debates about the method's internal consistency and optimal design.

We suggest that social scientists measuring IPV consider using both direct questions and the list experiment to test the likely magnitude of underreporting; use the double list experiment to both reduce variance in prevalence estimates, and to apply the internal consistency test proposed by Chuang et al. (2021); and test alternative methods and survey administration conditions that may minimize reporting bias (for example, testing the effect of enumerators more clearly explaining data confidentiality and ensuring interview privacy, or of incentivizing and monitoring empathetic enumeration). Further research could also explore the feasibility and reliability of alternative survey methods such as the ballot method, or of other non-self-reported indicators of IPV, such as brain imaging or stress hormones.

6. Data Availability

The data underlying this article were provided by the World Bank and Government of Rwanda by permission. Data will be shared on request to the author with permission of the World Bank and Government of Rwanda.

29 DHS data on IPV cover more than 60 developing countries and are cited in more than 5,000 papers on Google Scholar. If DHS IPV data are measured with substantial, systematic, and context-specific bias as this paper suggests is likely, this has significant implications for the use of such data.

Downloaded from https://academic.oup.com/wber/article/37/1/49/6883824 by International Monetary Fund / World Bank - IMF user on 11 September 2023